Introduction

Information Security

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Outline

- Part I: Few foundational terms and concepts
- Part II: Some use cases
- Part III: Lessons learned from use cases

Part I: Foundational Terms we Need

Can I give these terms for granted?

- Scalability
- Availability
- Single point of failure
- Bottleneck
- Integrity
- Confidentiality
- Privacy
- Fairness
- Auditability
- Client/Server model
- Peer-to-Peer model
- Churn

Maybe, we can focus on a subset

- Scalability
- Availability
- Single point of failure
- Bottleneck
- **Integrity**
- **Confidentiality**

- **Privacy**
- **Fairness**
- Auditability
- Client/Server model (**some issues only**)
- Peer-to-Peer model
- Churn → refers to the rate at which employees leave an organization. High information security churn causes a loss of institutional knowledge, disrupts security operations, and significantly increases an organization's vulnerability to breaches.

Integrity

Data integrity is the assurance that information remains accurate, complete, and consistent throughout its entire lifecycle

- Data not altered or destroyed in an unauthorized or accidental manner

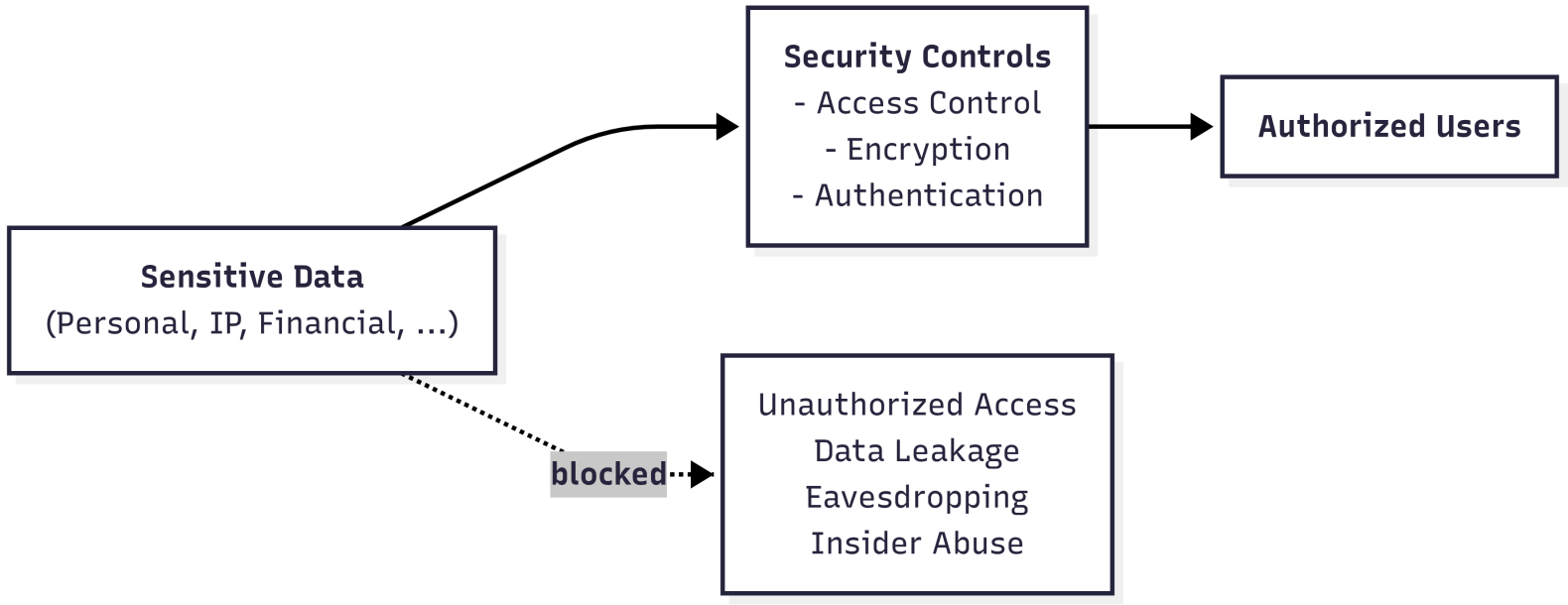
Integrity in ML

- **no data poisoning:** need to ensure training data is not altered
- **model integrity:** need to ensure model parameters are not altered

Confidentiality

Confidentiality ensures sensitive data is accessed only by authorized individuals

- Ensures that only authorized entities can access sensitive information
- Protects data from unauthorized disclosure or exposure



Privacy

Privacy is the right of individuals to exercise control over how their personal data is collected, processed, and shared, ensuring the confidentiality, integrity, and ethical use of their information

- **Privacy in Storage:** store only necessary data (**data minimization**), properly encrypted (**encryption at rest**)
- **Privacy in ML:** individuals not re-identified from model outputs (**membership inference protection**)

A Membership Inference Attack is a specific type of privacy attack in the field of machine learning.
- How the attack works: an attacker takes a record of an individual (e.g. Mario Rossi) and feeds it into the model. The attacker analyses the model's response (often by looking at the prediction confidence scores). Since models tend to respond much more accurately to data they have already seen during training than to entirely new data, the attacker can determine with a very high degree of certainty whether Mario Rossi's data was part of the training dataset.
- Why is this serious? Knowing that an individual is part of the dataset can, in itself, constitute a very serious privacy violation. For example: if the model in question was trained exclusively on patients with a specific rare disease, discovering that Mario Rossi's data was used for training automatically reveals that Mario Rossi has that disease. Protecting against this attack means ensuring that the model's behaviour is statistically identical whether an individual's data was included in the training or not.

Privacy vs Confidentiality

- **Confidentiality → access control**
- **Privacy → inference control** → Inference control analyzes and restricts the information that a user can deduce (infer) by querying a system or observing its outputs.

Normally, to protect sensitive data, you use access control: if a user is not authorized to view a specific file or piece of data, the system blocks them. The problem arises when a malicious user does not attempt to read the restricted data directly, but instead asks legitimate questions or analyzes the system's responses to guess (infer) what they are not supposed to see. Inference control is designed precisely to prevent this from happening, by monitoring or altering public results to ensure that no one can reconstruct the "secret" hidden behind the scenes

Fairness

Fairness ensures that information systems and algorithms operate without creating or reinforcing bias. It implies the equitable (fair) treatment of all users and data subjects, preventing discriminatory outcomes and ensuring balanced resource allocation across the network

- **Fairness in distributed systems**: fair resource allocation, laod balancing, latency equity
- **Fairness in ML**: bias mitigation, group fairness, algorithmic transparency

The problem of trust

Central Trusted Authority

In the traditional model of computer science (e.g., the classic client-server paradigm), security relied on the presence of a central entity that everyone trusted implicitly (the Central Trusted Authority). Examples: A bank's central server or a hospital's central database.

What we **trust**

- **services**
- **data sources**
- **data**

Under the protective wing of this central authority, traditional systems implicitly assumed that three key elements were always reliable and free from malicious manipulation:
- Services: it was taken for granted that software and algorithms performed calculations correctly, honestly, and without any anomalous behaviour.
- Data sources: it was assumed that those providing the data (e.g., official sensors, company employees, internal databases) were legitimate, authenticated, and non-malicious entities.
- Data: it was assumed that the data entered into the system was clean, accurate, and free from alterations inserted by third parties.

However, in modern systems, **trust cannot be assumed**

In today's systems, nothing can be trusted a priori:
- Unverified data sources: if data is collected collaboratively from thousands of smartphones belonging to unknown users (e.g., traffic data or GPS tracking), you can no longer assume that the data sources are trustworthy. A malicious user could send fake data to compromise the system (Integrity breach).
- The AI supply chain: When training a machine learning model, data is often labelled by third-party companies or models are updated in a decentralized manner (as in Federated Learning). This means an attacker can corrupt the underlying data or send malicious updates directly during training (Data Poisoning and Label Flipping attacks).
- The intrinsic vulnerability of aggregated data: even if the sources appear anonymous, the combination and aggregation of massive amounts of modern data (using OSINT) allows attackers to perform advanced correlations, violating privacy through inference (as demonstrated by the historic Netflix dataset).

Two Fundamental Objectives

Systems should be analyzed along two axes (~~**besides**~~ in addition to **performance**):

- **Trustworthiness**
  → Can we rely on data, models, and system behavior?

In modern systems, data comes from unverified sources (e.g., users' smartphones, IoT sensors, the web), and decision-making models (ML) are complex black boxes. Ensuring trustworthiness means implementing mechanisms to ensure that:
- The data is intact: No one has manipulated or altered it at the source (preventing attacks such as data poisoning).
- The models are robust: The algorithm makes correct decisions and is not easily fooled by malicious inputs (preventing evasion attacks or adversarial examples).
- The behavior is predictable: The system does exactly what is expected, without dangerous emergent behaviors or exploitable logical flaws.

- **Privacy**
  → Can we use data without exposing sensitive information?

Modern systems have a hungry need for data to function properly (the more clean data there is, the more AI learns). The Privacy axis focuses on ethical and legal constraints: how can we extract value, statistics, and patterns from this massive amount of data without ever violating the anonymity or rights of the individuals who generated that data? As we saw earlier, this requires controlling inferences (preventing the output from being traced back to an individual).

These are **not independent** and often introduce trade-offs

Increasing security in one area can lower the security of the other. You cannot have the best of both worlds without paying a price. Here are the three main trade-offs between Trustworthiness and Privacy:
1) The Control Trade-off (Verification vs. Anonymity)
To increase Trustworthiness: If I want to be 100% sure that a data source is reliable and isn't sending false data (e.g., fake reviews, altered GPS coordinates), the most logical approach is to identify, track, and audit the data sender. Knowing exactly who sent what destroys the user's privacy. Conversely, if I guarantee total anonymity to protect privacy, I open the door to malicious actors who can send junk data without being detected, undermining the system's trustworthiness.
2) The Trade-off of Statistical Noise (Differential Privacy vs. Accuracy)
To Increase Privacy: To protect ML models from privacy-violating attacks (such as membership inference), we use Differential Privacy, which introduces "random noise" into the data or the model's calculations. If I introduce too much noise, the model becomes less accurate, less stable, and potentially less robust. If a medical model misdiagnoses a patient because we introduced too much noise to protect patient privacy, the system is no longer trustworthy.
3) The Transparency Trade-off (Explainability vs. Vulnerability)
To increase Trustworthiness: A system is considered trustworthy if it is transparent: that is, if the user or certifier can understand why the model made a certain decision (Explainable AI). The more details and explanations the model provides about its internal workings or the reasoning behind its response, the more information it is giving away to an attacker, who can use those explanations to reconstruct sensitive training data (Model Inversion) or to bypass the model (Evasion).

Trust vs Privacy: A Tension

Improving one objective may weaken the other

Examples:

1. Logging improves **auditability** but may reduce **privacy**
   - Data sharing improves **utility** but increases **exposure risk**
2. Differential Privacy improves privacy but reduces model accuracy

Does the AI revolution change the overall picture?

In terms of trustworthiness, privacy, scalability, fairness, auditability, etc.

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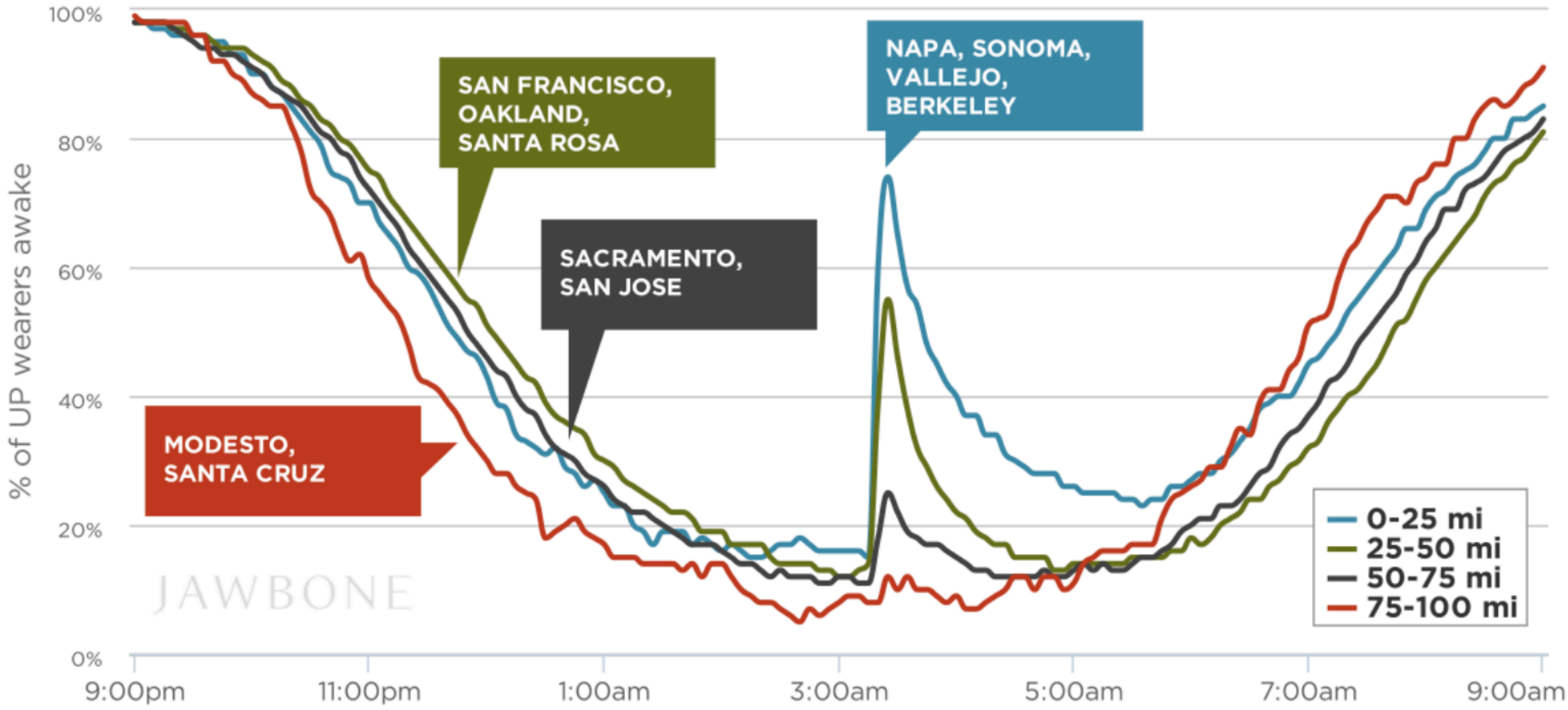
In terms of trustworthiness, privacy, scalability, fairness, auditability, etc.

Trustworthy AI

Designing and ensuring Trustworthy AI means creating Artificial Intelligence systems that humanity can rely on safely and ethically, even in the presence of hostile environments or malicious attackers. A Trustworthy AI must simultaneously meet four fundamental requirements:

- Adversarial Robustness: the model must be mathematically and statistically robust against manipulation attempts. It must be able to withstand attacks both during the training phase (using Robust Aggregation Rules to neutralize poisoned updates) and during the inference phase (using Adversarial Training to avoid being fooled by adversarial examples).
- Certified Mathematical Privacy: a Trustworthy AI must incorporate advanced protections such as Differential Privacy, introducing calibrated noise into calculations to mathematically guarantee that the model's output never reveals whether a single individual's record was used or not.
- Data Supply Chain Governance and Control: since AI is vulnerable to corrupted data, ensuring reliability requires strict control over the data lifecycle. For example, risks associated with external suppliers labelling data can be mitigated by implementing redundant checks (e.g., having the same record labelled by multiple independent annotators and measuring the degree of agreement).
- Emergent Security in Distributed Systems: when multiple AI agents interact with one another in a decentralized manner, the system becomes dynamic and unpredictable. In this context, a Trustworthy AI must be able to manage trust dynamically and rely on engineered network structures (Topology) to prevent the compromise of a single agent from causing the collapse or misinformation of the entire network.

The landscape has changed



The more data you collect from various sources, the more your knowledge grows. A single piece of public data on its own means nothing, but millions of data points together create value.
Correlation & Patterns: By combining different data, algorithms and analysts can identify stable correlations and recurring patterns that were previously invisible.
Insights: From these correlations, deep insights, predictive models, and strategic decisions are derived.

The landscape has changed

Open Source Intelligence (OSINT)

The more data is publicly available, the larger the attack surface becomes, and the more people (or organizations) are exposed.
Inference: An attacker can use seemingly harmless pieces of information and, by combining them, deduce (infer) a secret that was never publicly disclosed.
Re-identification: Even if a company publishes an anonymized dataset, an attacker can take that dataset and cross-reference it with other OSINT sources. Through this correlation, the attacker can uncover the real identity behind that anonymous data.
Hidden sensitivity: Data that today seems trivial and non-sensitive (e.g., your running routes tracked by an app), when aggregated with other data, can reveal highly sensitive information.

More data → More knowledge

- Correlation
- Patterns
- Insights

Aggregation is power

More data → More exposure

- Inference
- Re-identification
- Hidden sensitivity

Aggregation is risk

Real-world example

Netflix Prize Dataset (2006)

- “Anonymized” movie ratings released
- No names, no direct identifiers

↓

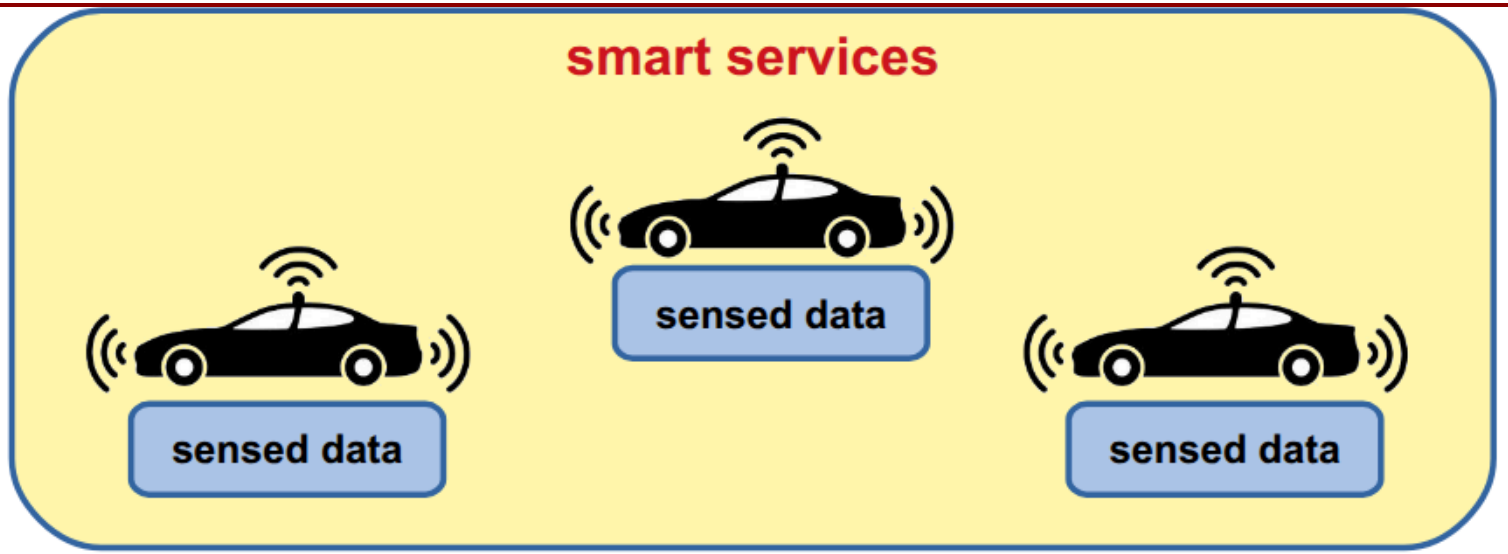
- Cross-referenced with IMDb ratings → The Netflix dataset was cross-referenced with public data extracted from IMDb (Internet Movie Database), a well-known online platform (and source of OSINT) where users review and rate movies in a completely public manner, signing in with their first and last names or a traceable nickname.
- Users re-identified

Part II: Some Use Cases

Use Case #1: Crowdsensing Applications

(in Decentralized Systems)

Smart Transportation Systems



Smart Transportation Systems

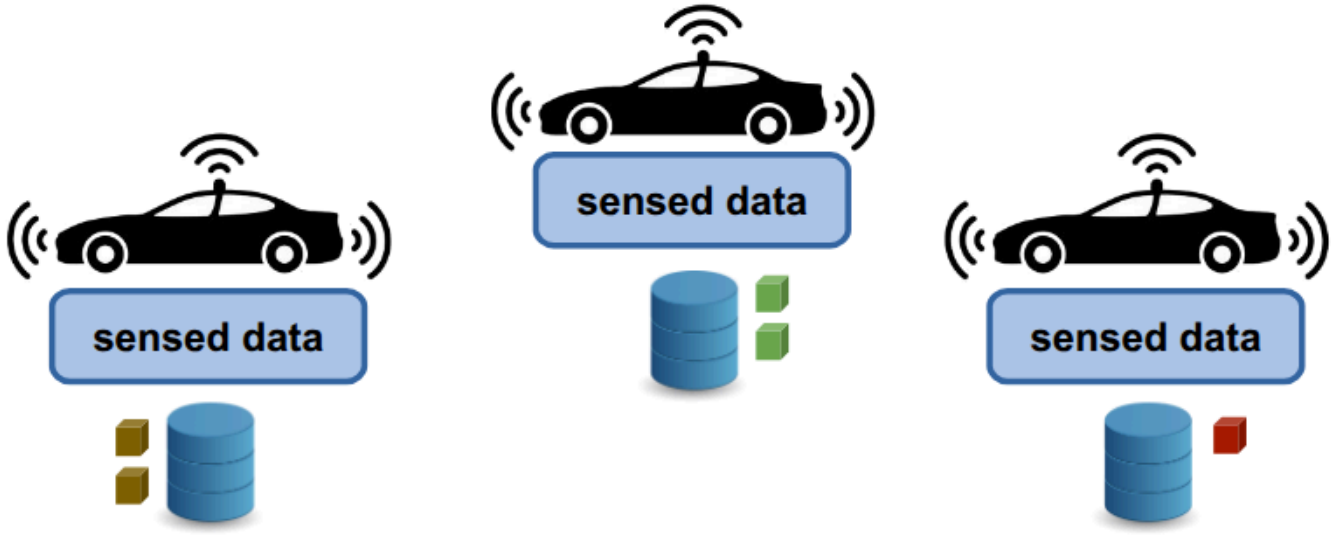
The system implements **crowdsensing-as-a-Service**

Crowdsensing is a paradigm in which a multitude of ordinary users, using sensors built into their everyday mobile devices (such as smartphones, smartwatches, or in-car computers equipped with GPS, accelerometers, cameras, etc.), collect and share data useful for monitoring a phenomenon of common interest. When this paradigm becomes as-a-Service, it means that a decentralized technological infrastructure is established to continuously collect, aggregate, and process this mass data in order to resell it or offer it in the form of application services to users or companies.

Data used to provide:

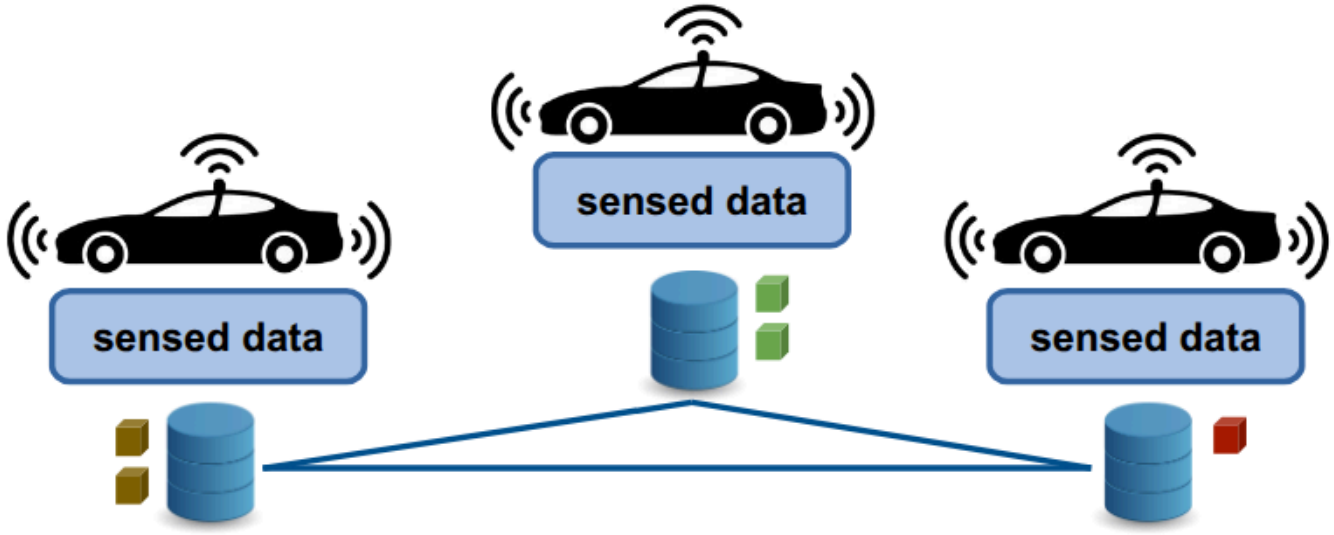
- Traffic analysis
- Route optimization
- Safety alerts (e.g., crash, wrong-way driving, weather conditions)

Smart Transportation Systems



Personal data

Smart Transportation Systems



Data storage

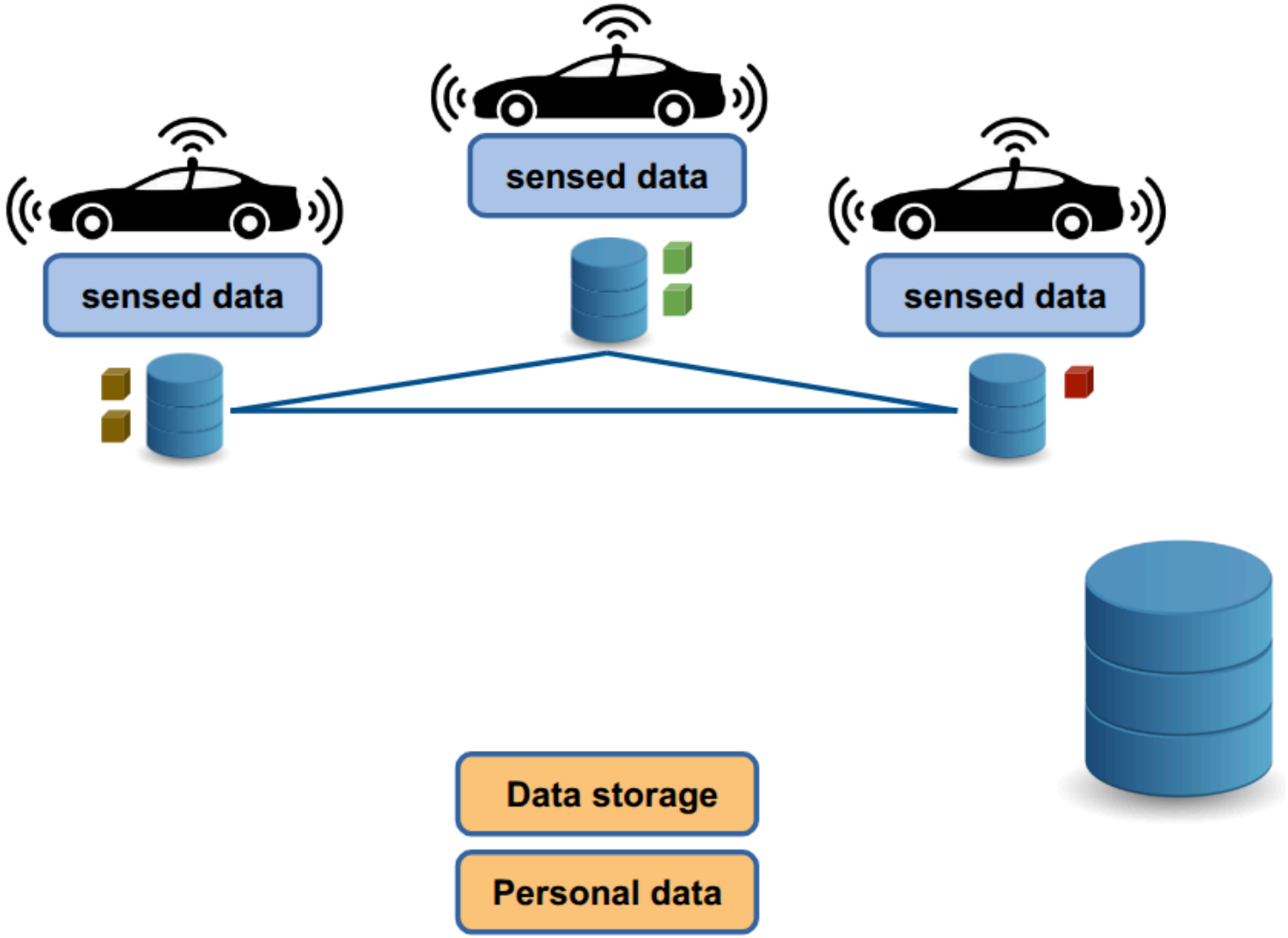
Personal data

Security Objectives

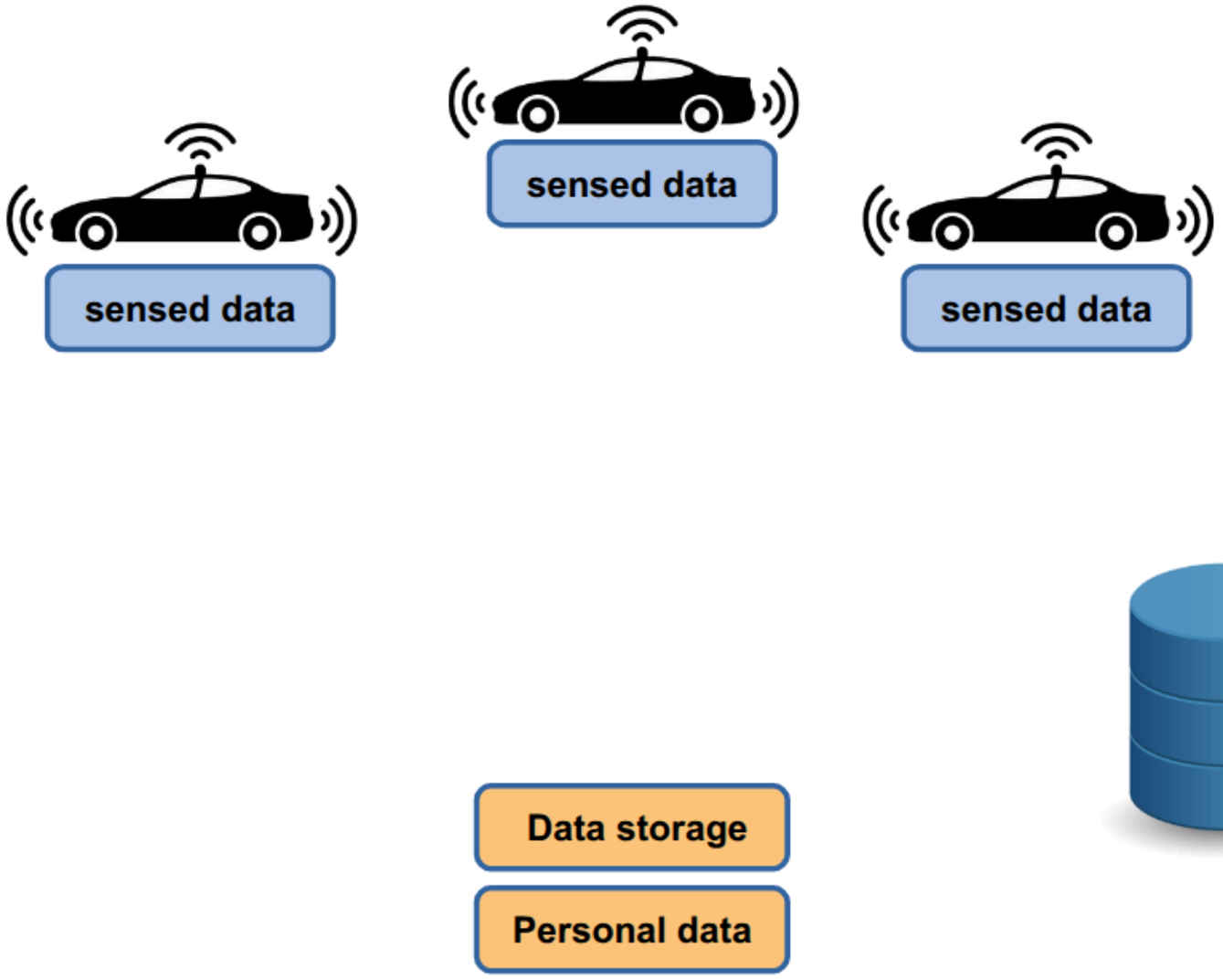
The system must guarantee:

- **Integrity** → Data cannot be altered undetectably
- **Confidentiality** → Unauthorized parties cannot access raw data
- **Availability** → Data remains retrievable when needed
- **Authenticity** → Data provenance is verifiable
- **User Sovereignty** → Data subjects retain control

Opt1: central entity maintains crowdsourced data



Opt1: central entity maintains crowdsourced data



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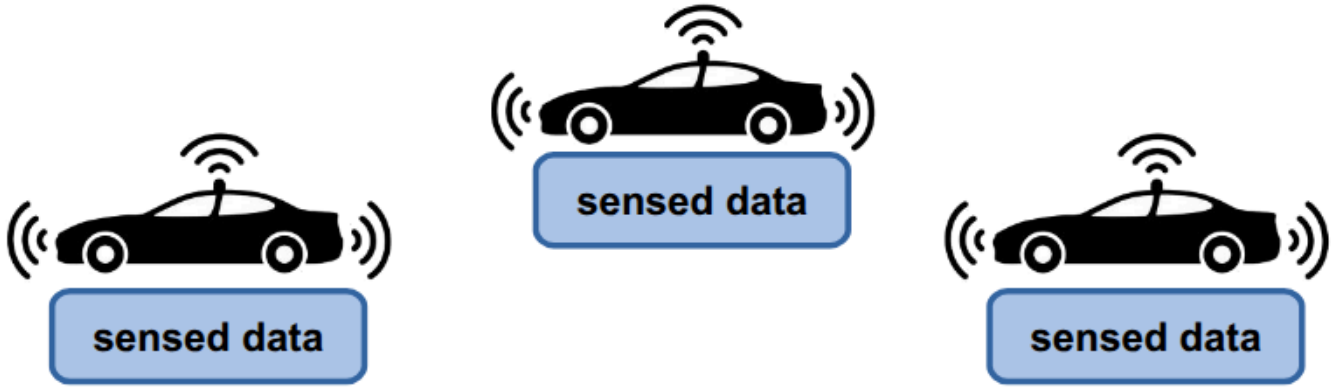


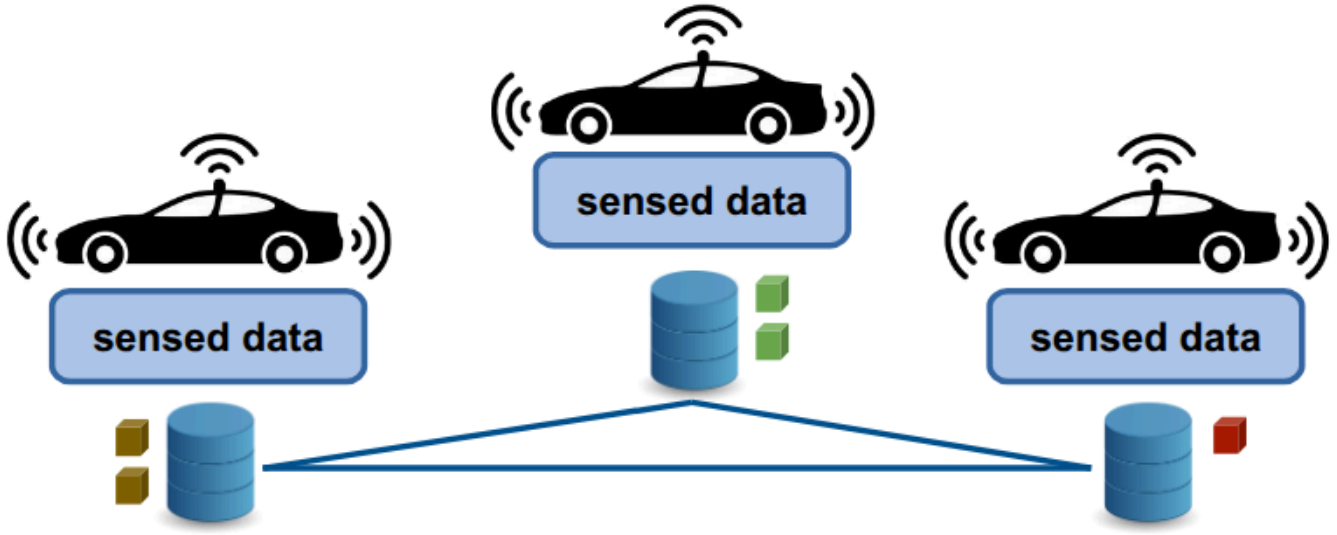
Option 1 cannot be implemented because it does not respect the principle of user sovereignty

Users lose the sovereignty over their data
- data controller gains all the data
- data controller can alter the data
- users must rely on the controller

Data storage

Personal data

Opt2: keep data locally - distribute upon request



Data storage

Personal data

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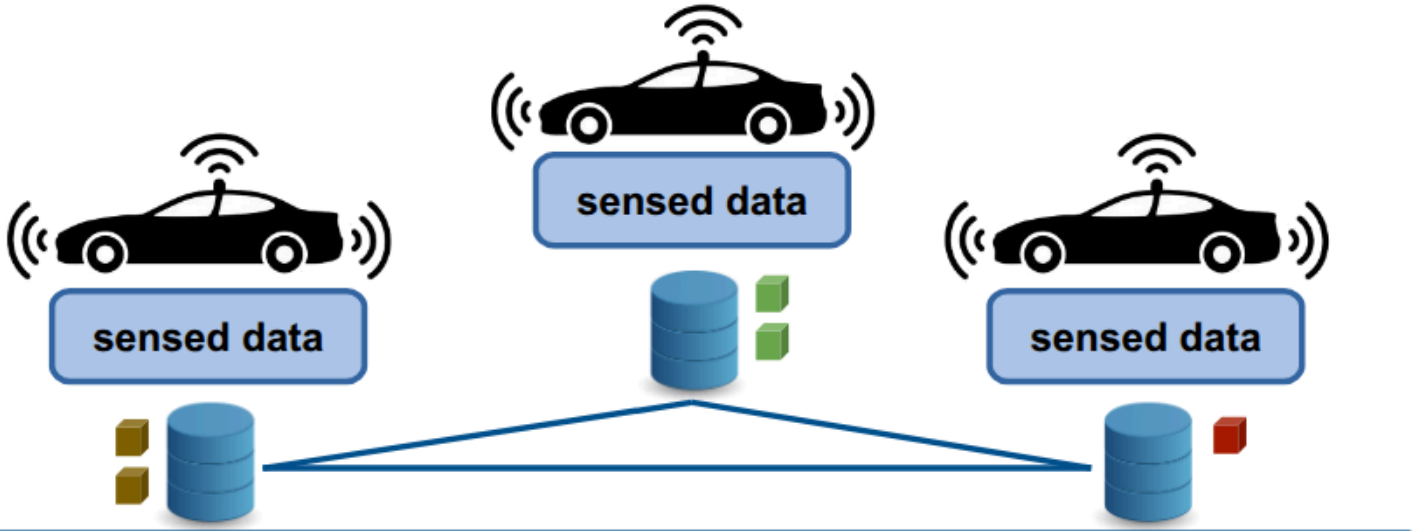


Pros
- you maintain your own data

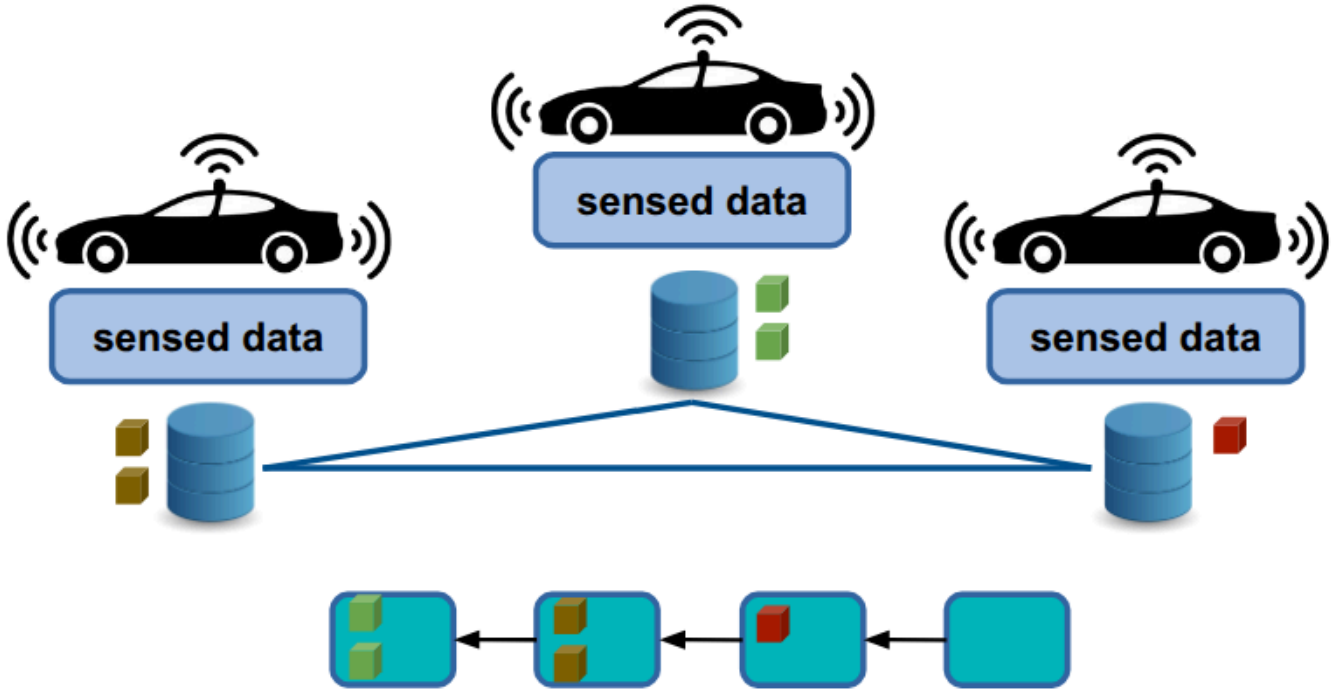
Cons
- not practical, you need to be always reachable
- storage, computation, communication capabilities needed

Data storage

Personal data

Opt3: use a ledger to register data

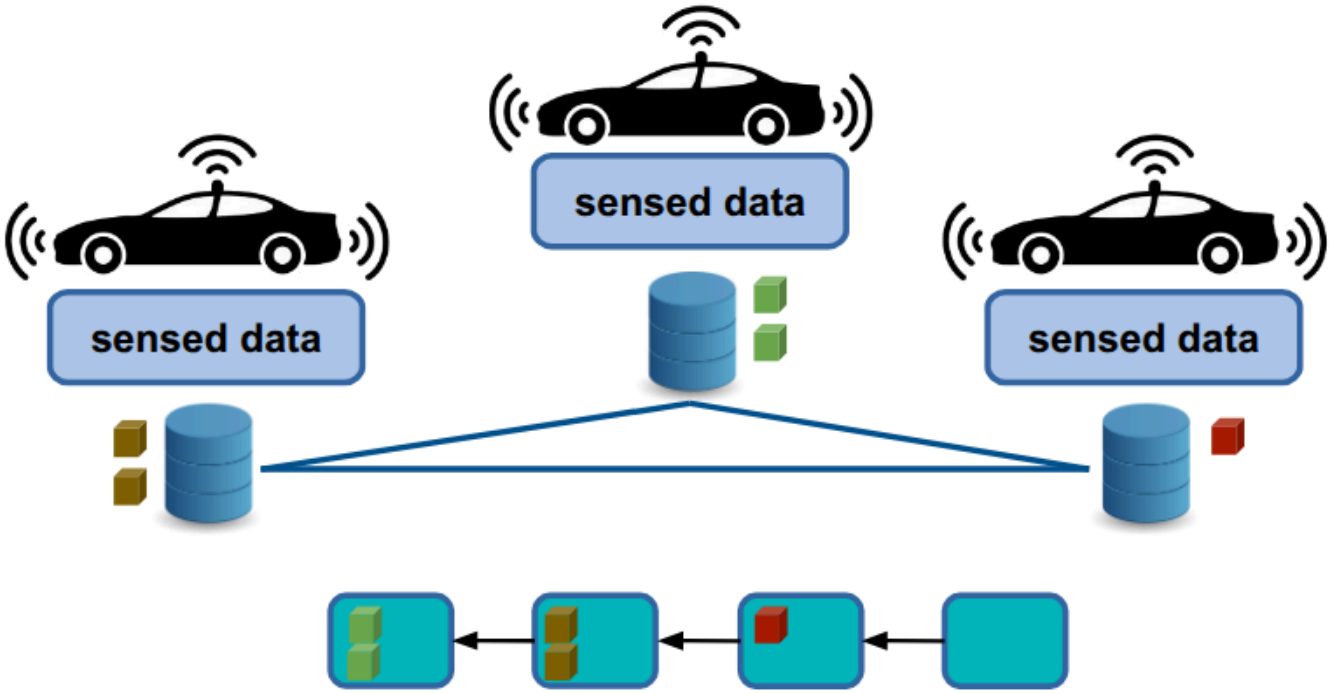
A ledger is a shared, decentralized digital record of data. Its key feature is that it does not live on a single company server, but is replicated and synchronized across a network of many independent computers (nodes). A ledger has two security-critical properties:
- Append-only: New data can only be added to the end; nothing that has already been written can be modified or deleted.
- Immutability (Anti-Tampering): Thanks to encryption, once data is recorded on the ledger, it becomes mathematically impossible for anyone (including the service provider or a hacker) to alter or falsify it retroactively without being detected.



Data storage

Personal data

Opt3: use a ledger to register data



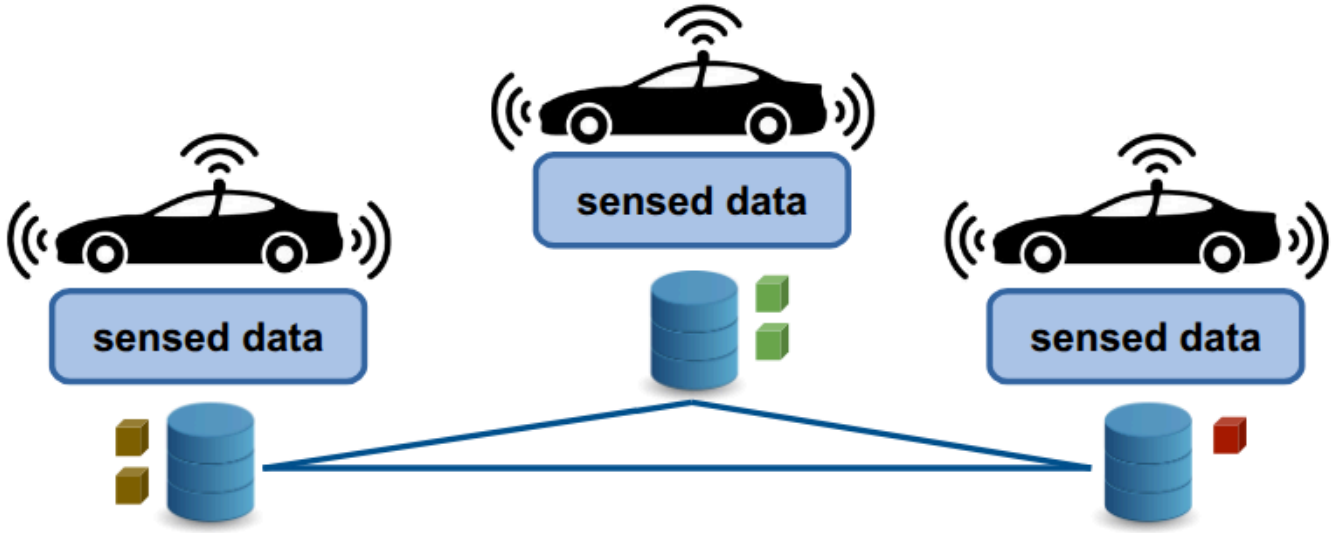
Pros:
- rather simple
- data integrity
- traceability

Data storage

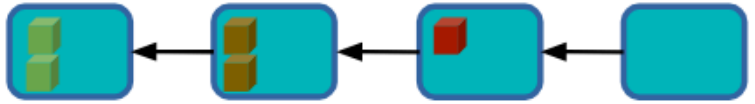
Personal data

Opt3: use a ledger to register data

- Simple: instead of implementing complex synchronization protocols or having to build custom distributed databases from scratch, the architecture relies on an existing, standardized ledger infrastructure. The system simply needs to send transactions in append to the ledger.
- Data Integrity: since modern ledgers are structurally immutable and use cryptography, once the data is written to the ledger, it can no longer be altered, replaced, or retroactively deleted by anyone. This neutralises attacks at their source where an attacker attempts to tamper with or falsify historical mobility data (hash substitution attempts).
- Traceability (Verifiability): every single data entry in the ledger receives an immutable timestamp (verifiable timestamp) and is digitally signed. This provides total and transparent traceability (perfect for auditability): anyone with the necessary permissions can verify the exact chronology of the data entered from its source, ensuring the authenticity of its origin.



- Only suitable for small amounts of data: writing information “inside” the blocks of a distributed ledger requires computation, network consensus, and storage space duplicated across many nodes. Consequently, storing large amounts of data (such as continuous GPS tracks or multimedia data) is economically unsustainable and technically prohibitive.
- Latency: a traditional database responds in fractions of a millisecond. A decentralized ledger, however, requires network nodes to reach algorithmic consensus to confirm a transaction. This introduces delays (latency). If thousands of vehicles attempt to record data simultaneously, there is a risk of ledger congestion: transactions remain in the queue, wait times skyrocket, and the entire real-time application risks freezing (availability attack).
- No right to be forgotten or to rectification: it is the greatest trade-off between integrity and privacy. Under current regulations (such as the GDPR), users have the legal right to request the deletion of their personal data (“ right to be forgotten”). However, the fundamental property of the Ledger is perpetual immutability (you can only add data to the end; nothing is ever deleted). If sensitive or identifying data ends up on the ledger by mistake, it will remain there forever, violating the principle of User Sovereignty (the user’s control over their own data).



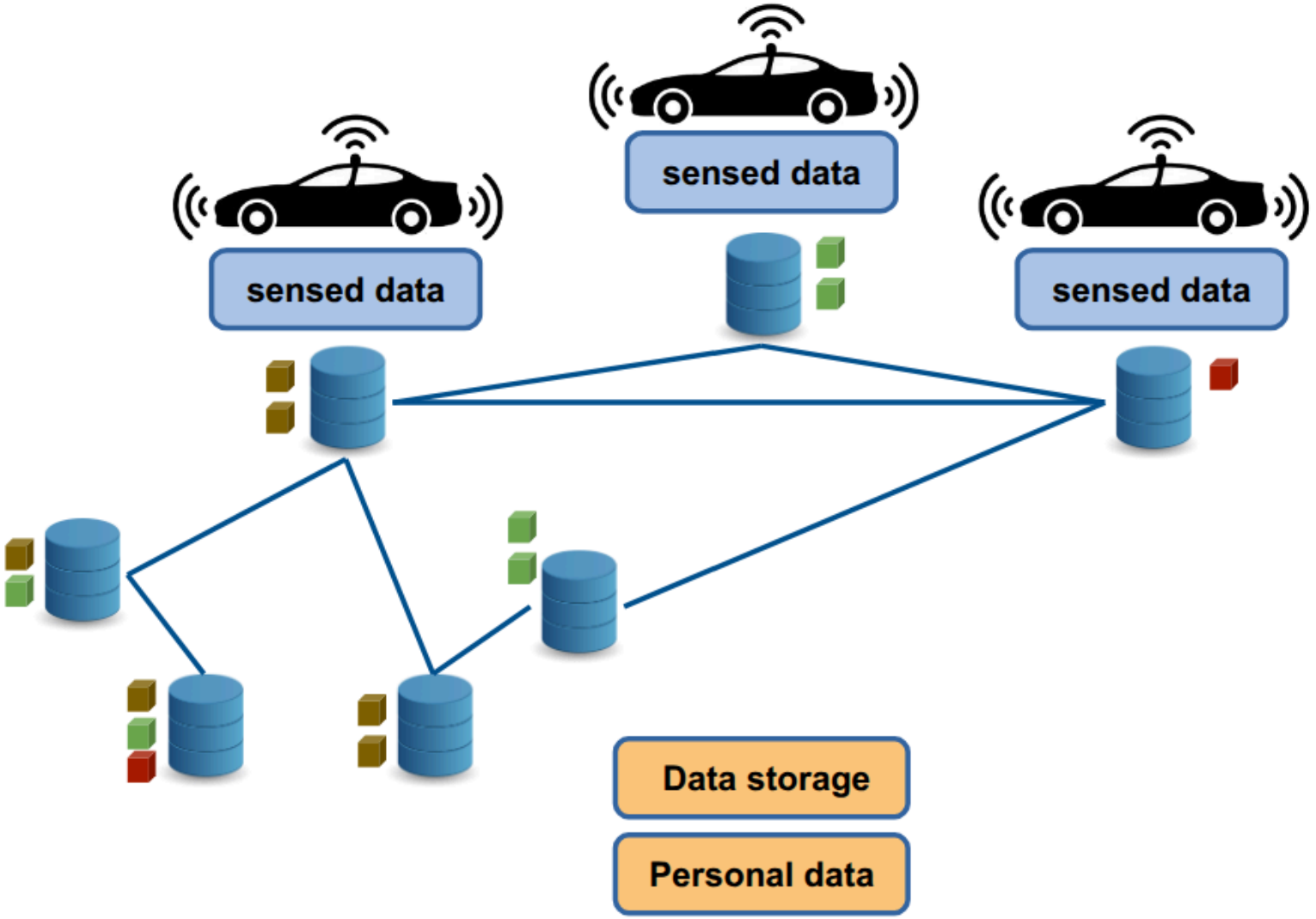
Pros:
- rather simple
- data integrity
- traceability

Data storage

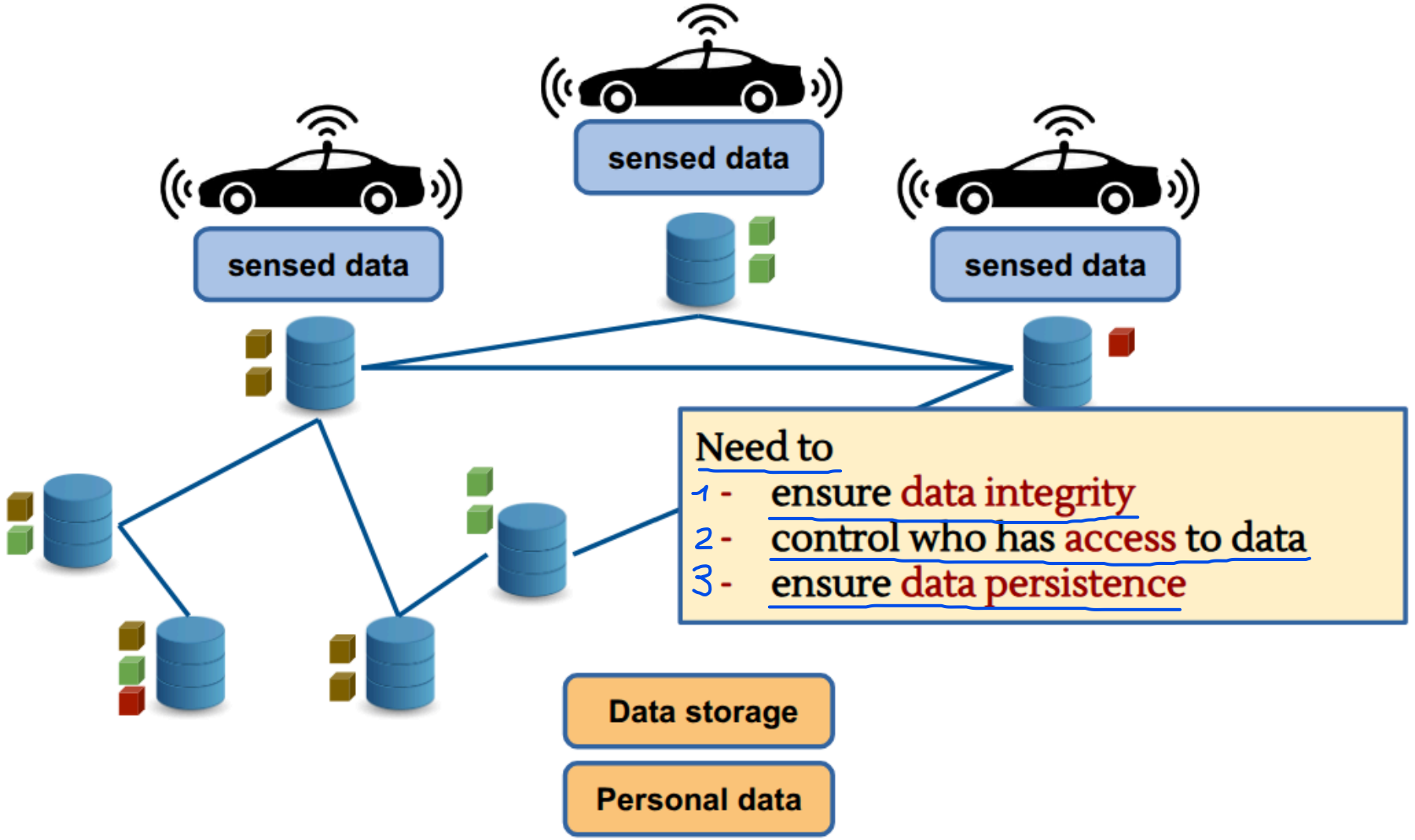
Personal data

Cons:
- ok with small sized data, only
- no right to be forgotten/rectified
- latencies

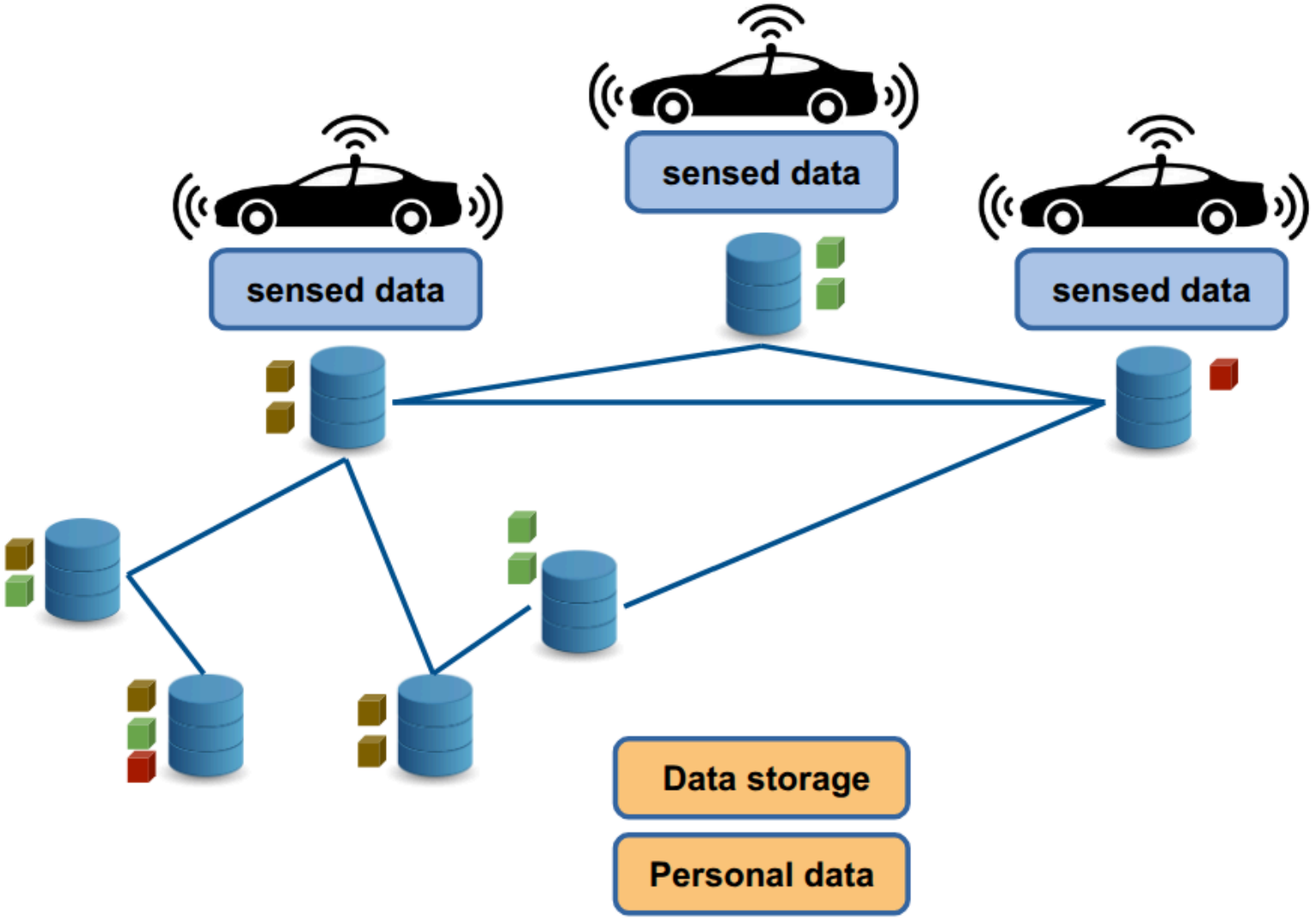
Opt4: use decentralized file systems (DFS)



Opt4: use decentralized file systems



Data Integrity

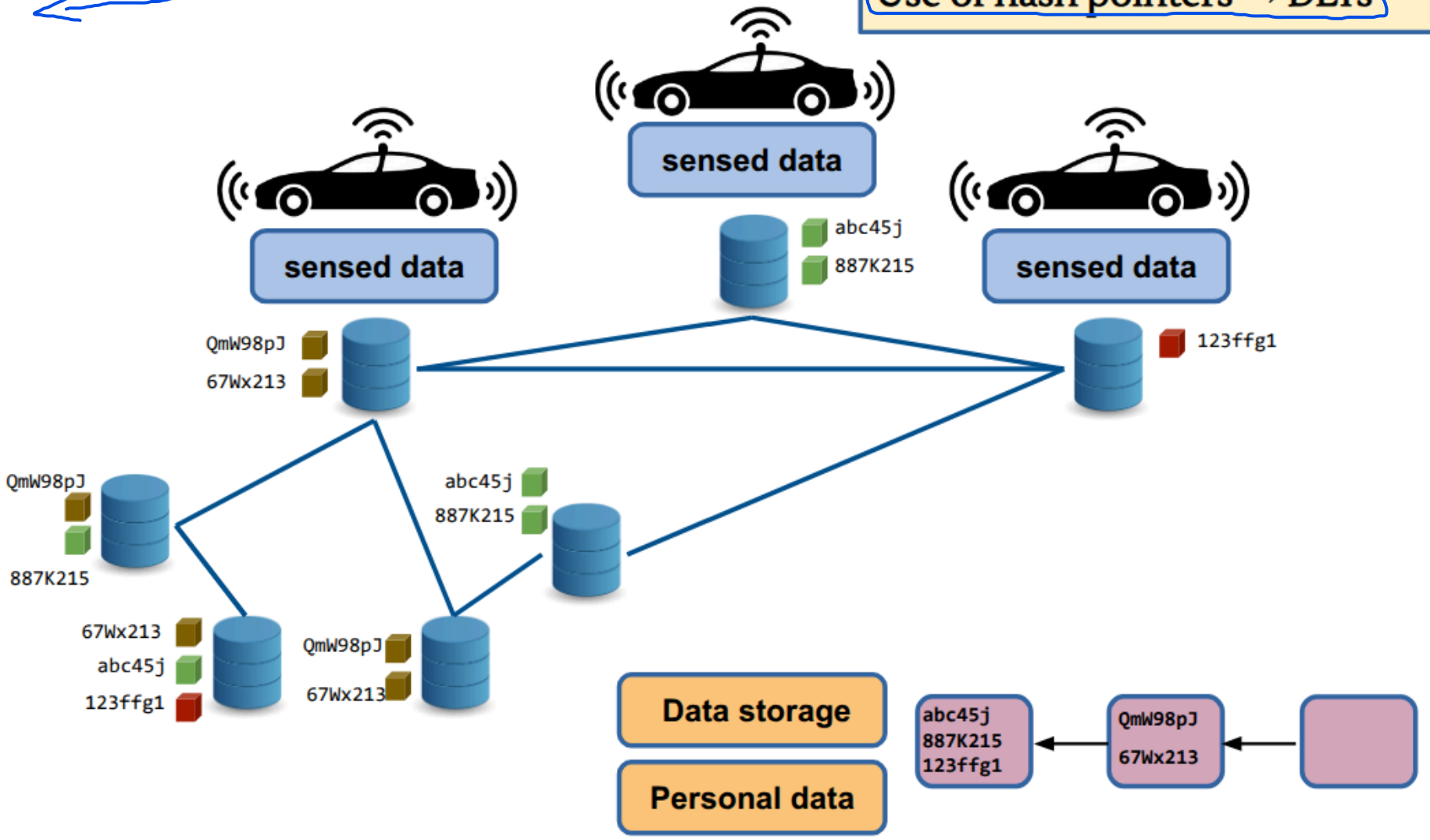


Data Integrity

Content based addressing
(instead of location based)

Use of hash pointers → DLTs

1. Content-based addressing: In decentralized file systems, a file is not located based on its geographic location or the server, but based on its content. The system takes the file's content, calculates its cryptographic hash, and uses that unique string as the address to retrieve it. Consequently, the file's address is the file's digital fingerprint. If you search for that code, the DFS network will find any node that possesses an exact copy of that content.
2. Use of hash pointers -> DLTs As we saw when analyzing Opt3, storing large amounts of mobility data directly within a DLT is impossible due to cost and congestion issues. A hash pointer is a pointer (an address) composed of the hash of the stored file. It serves a dual purpose: it tells you what you're looking for (content-based addressing) and provides mathematical certainty that the content hasn't been altered (because if it were altered, the hash would change). The connection to the DLT: The user uploads the large file to the DFS network, obtains the hash pointer, and writes only this small pointer to the distributed ledger (DLT).

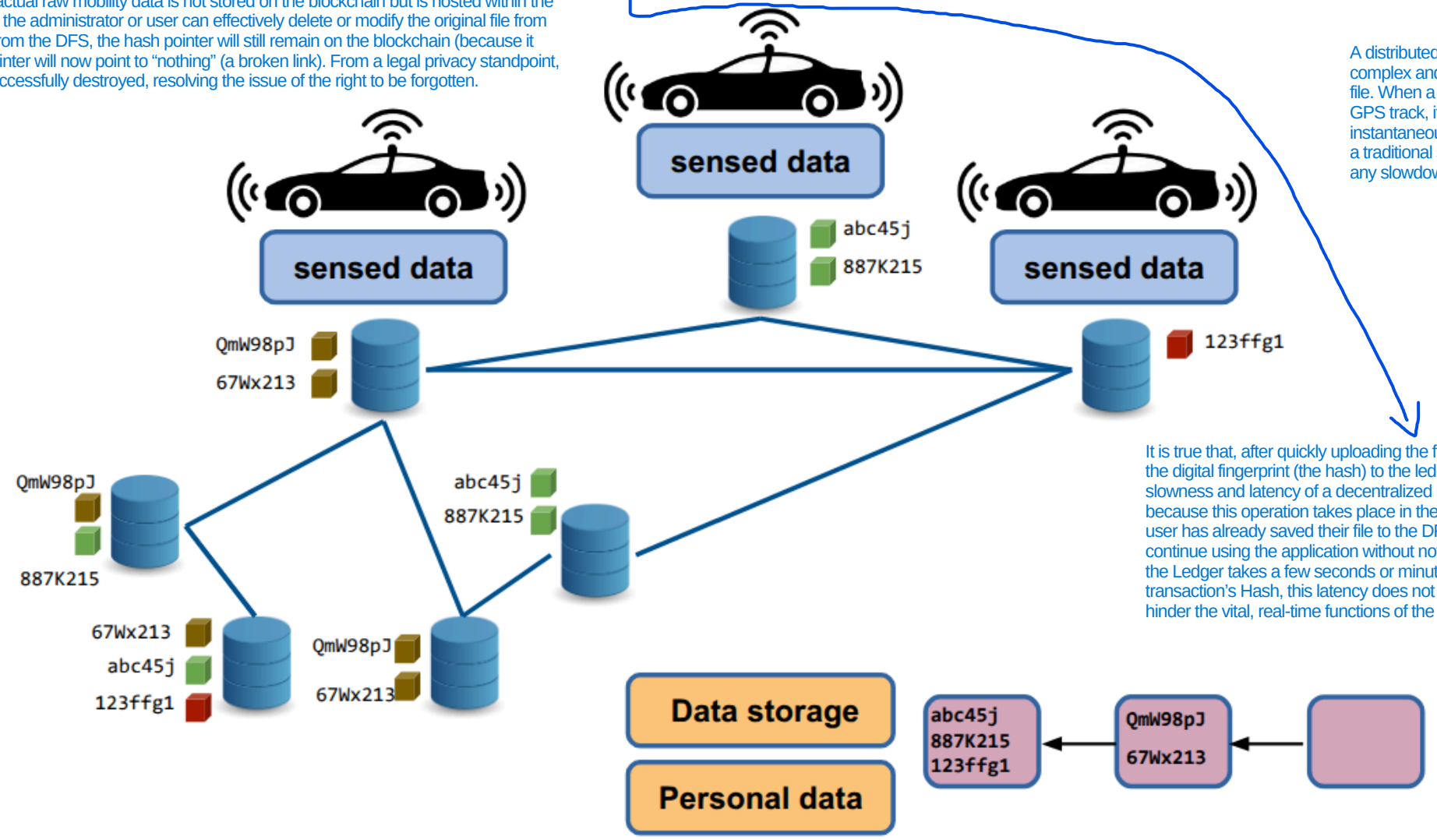


Data Integrity

- Possible to remove/modify data
- Fast data upload to DFS
- Latencies to upload hashes less problematic

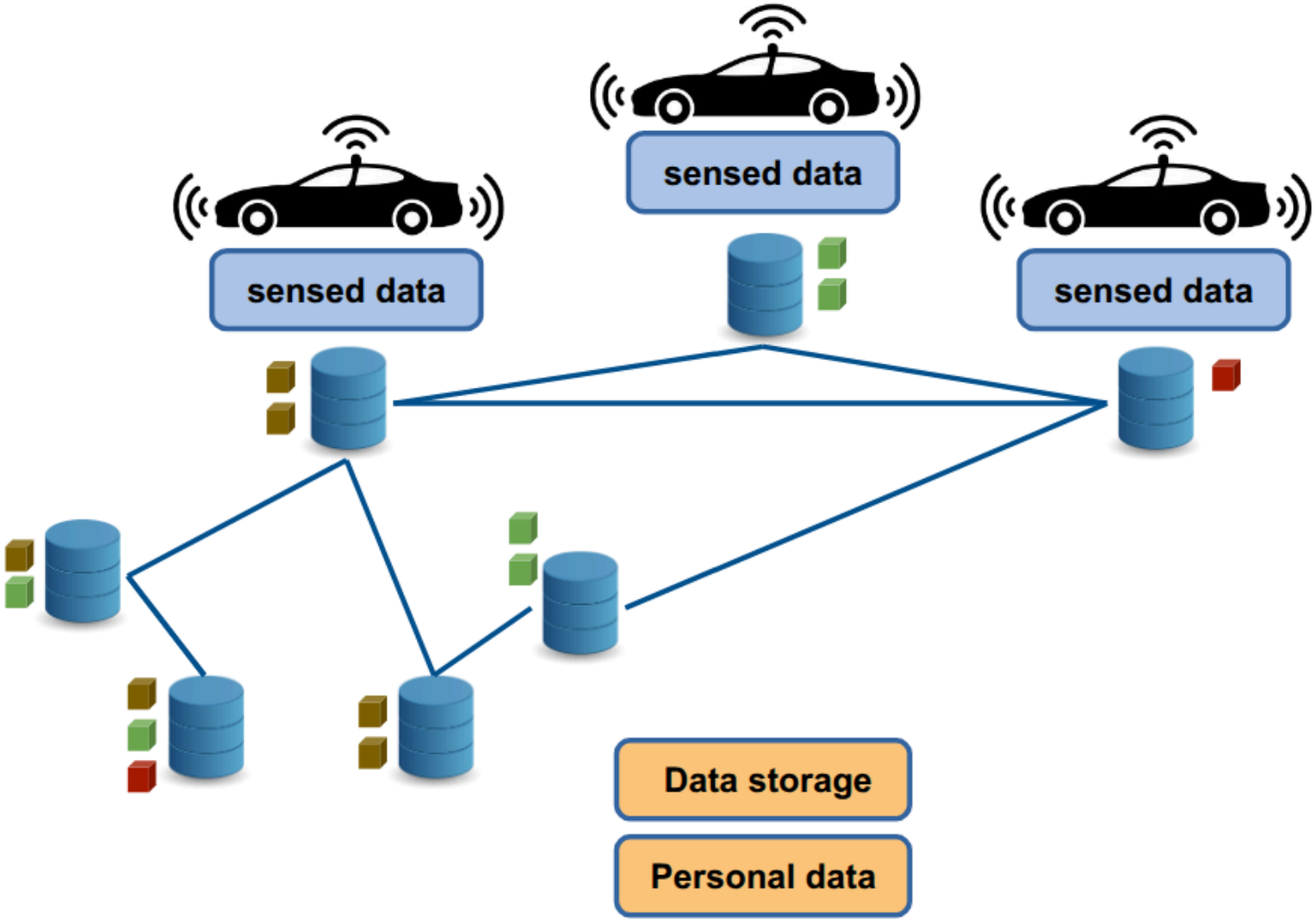
Since, in this architecture, the actual raw mobility data is not stored on the blockchain but is hosted within the DFS (Distributed File System), the administrator or user can effectively delete or modify the original file from the DFS. If you delete the file from the DFS, the hash pointer will still remain on the blockchain (because it cannot be deleted), but that pointer will now point to "nothing" (a broken link). From a legal privacy standpoint, the personal data has been successfully destroyed, resolving the issue of the right to be forgotten.

A distributed file system (DFS) does not require complex and slow consensus algorithms to accept a file. When a smartphone or car app needs to save a GPS track, it sends it to the DFS; this process is instantaneous and extremely fast, almost on par with a traditional server. The user does not experience any slowdowns in the app.

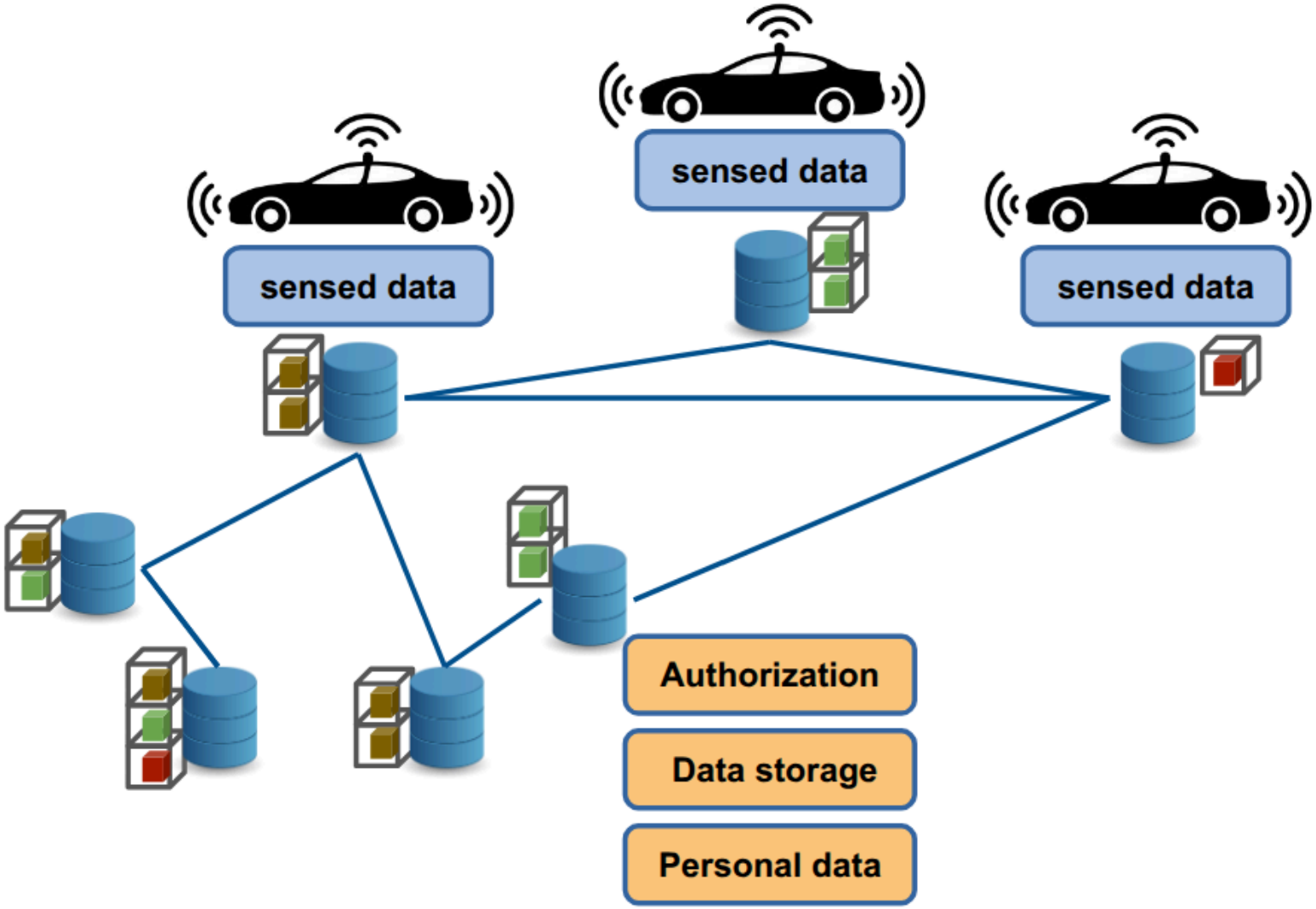


It is true that, after quickly uploading the file to the DFS, the system must still send the digital fingerprint (the hash) to the ledger to certify it, experiencing the typical slowness and latency of a decentralized network. This is less problematic because this operation takes place in the background and is asynchronous. The user has already saved their file to the DFS in just a few milliseconds and can continue using the application without noticing any freezes or slowdowns. Even if the Ledger takes a few seconds or minutes to confirm and "anchor" the transaction's Hash, this latency does not ruin the user experience and does not hinder the vital, real-time functions of the Smart Transportation System.

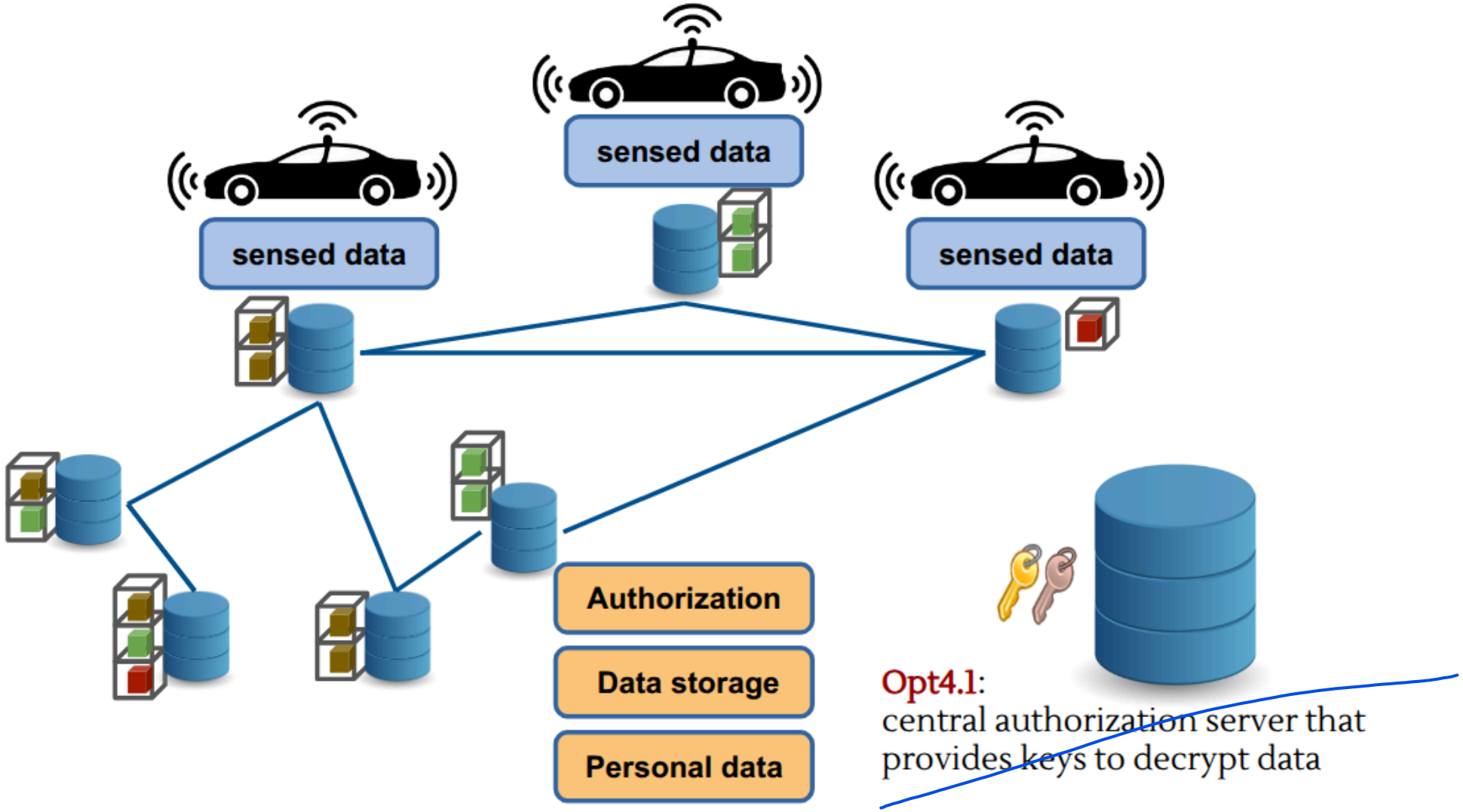
Access Control?



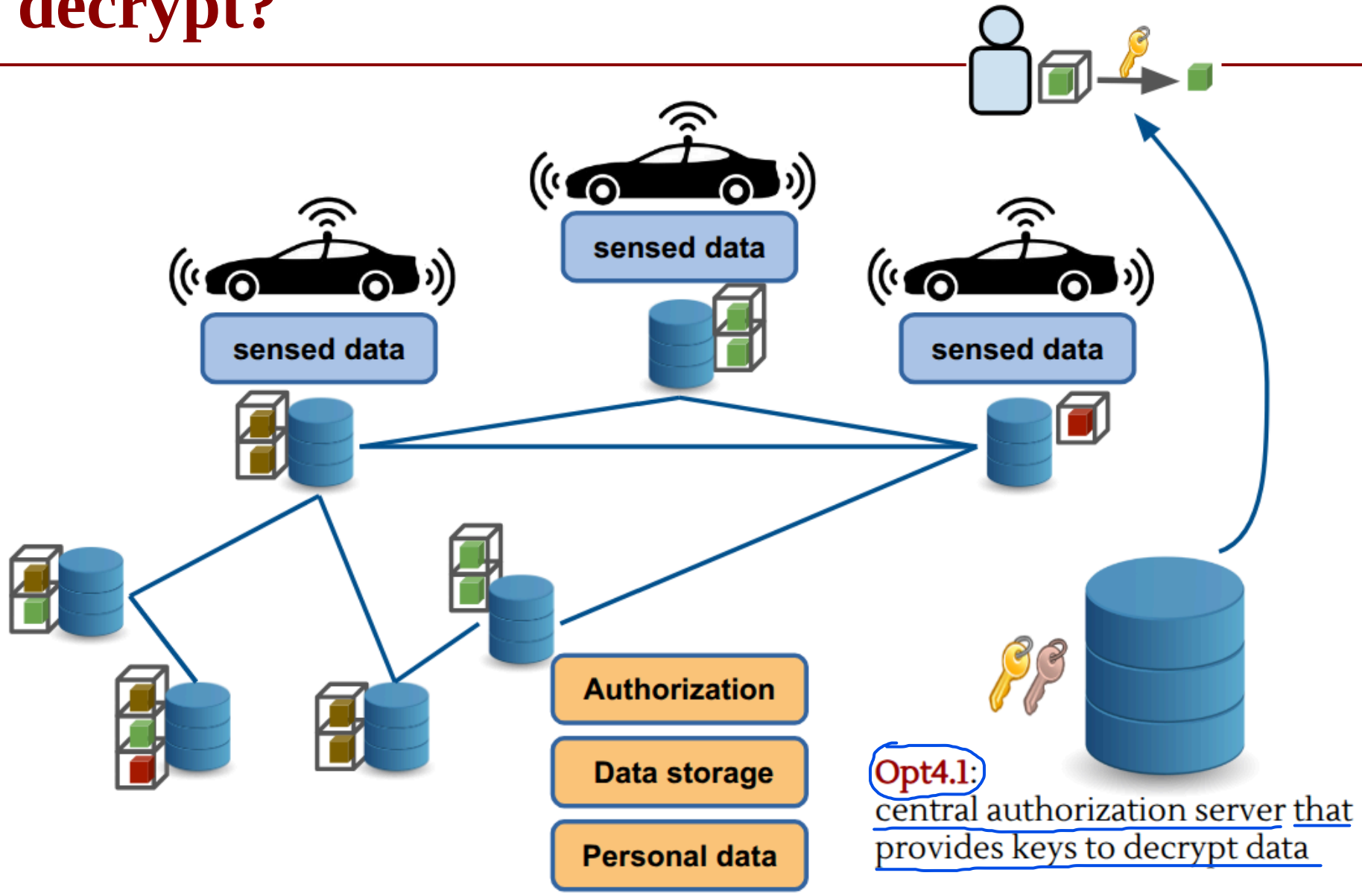
Access Control: DFS + Encryption



How to decrypt?

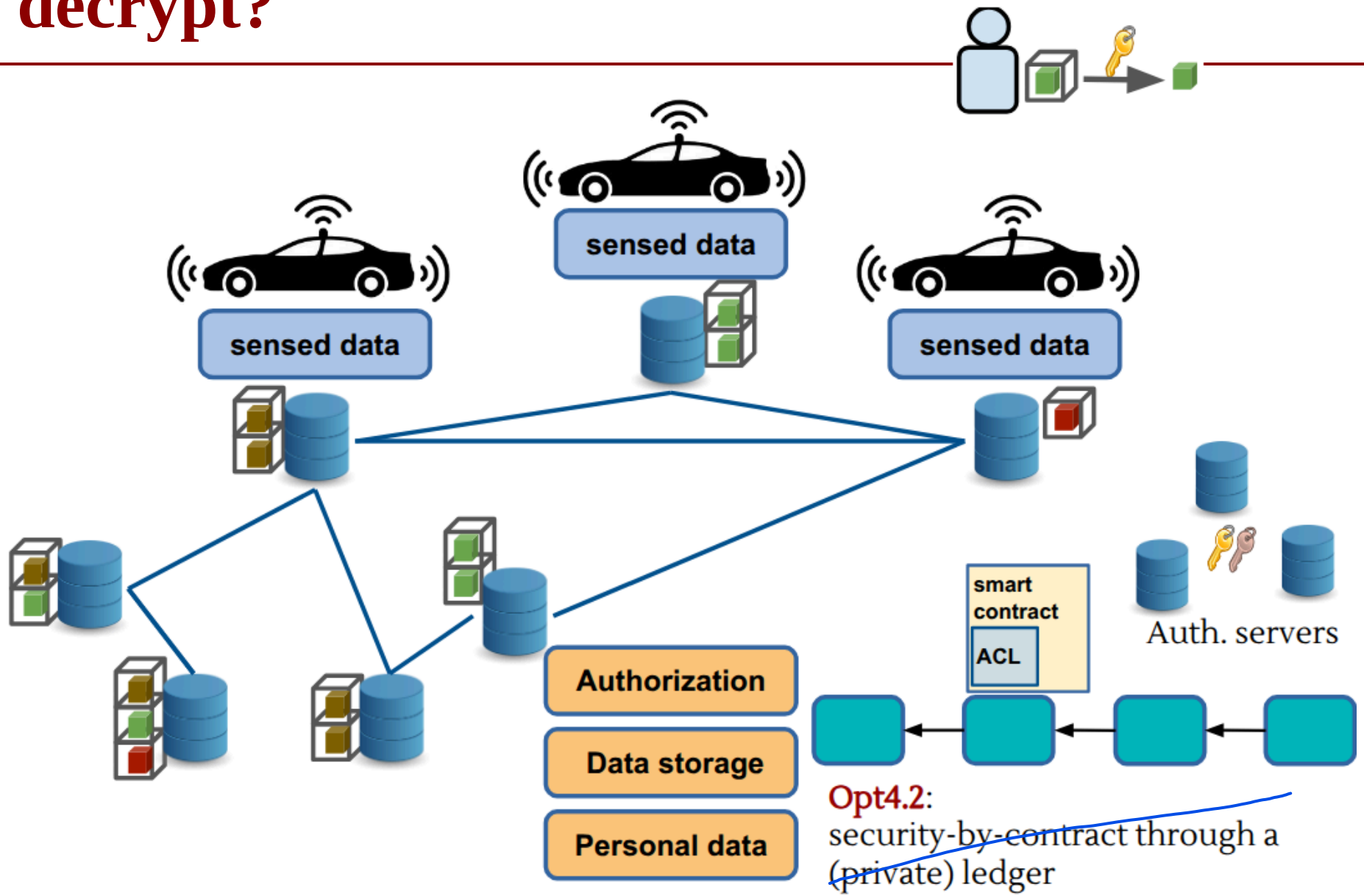


How to decrypt?



Opt4.1:
central authorization server that provides keys to decrypt data

How to decrypt?

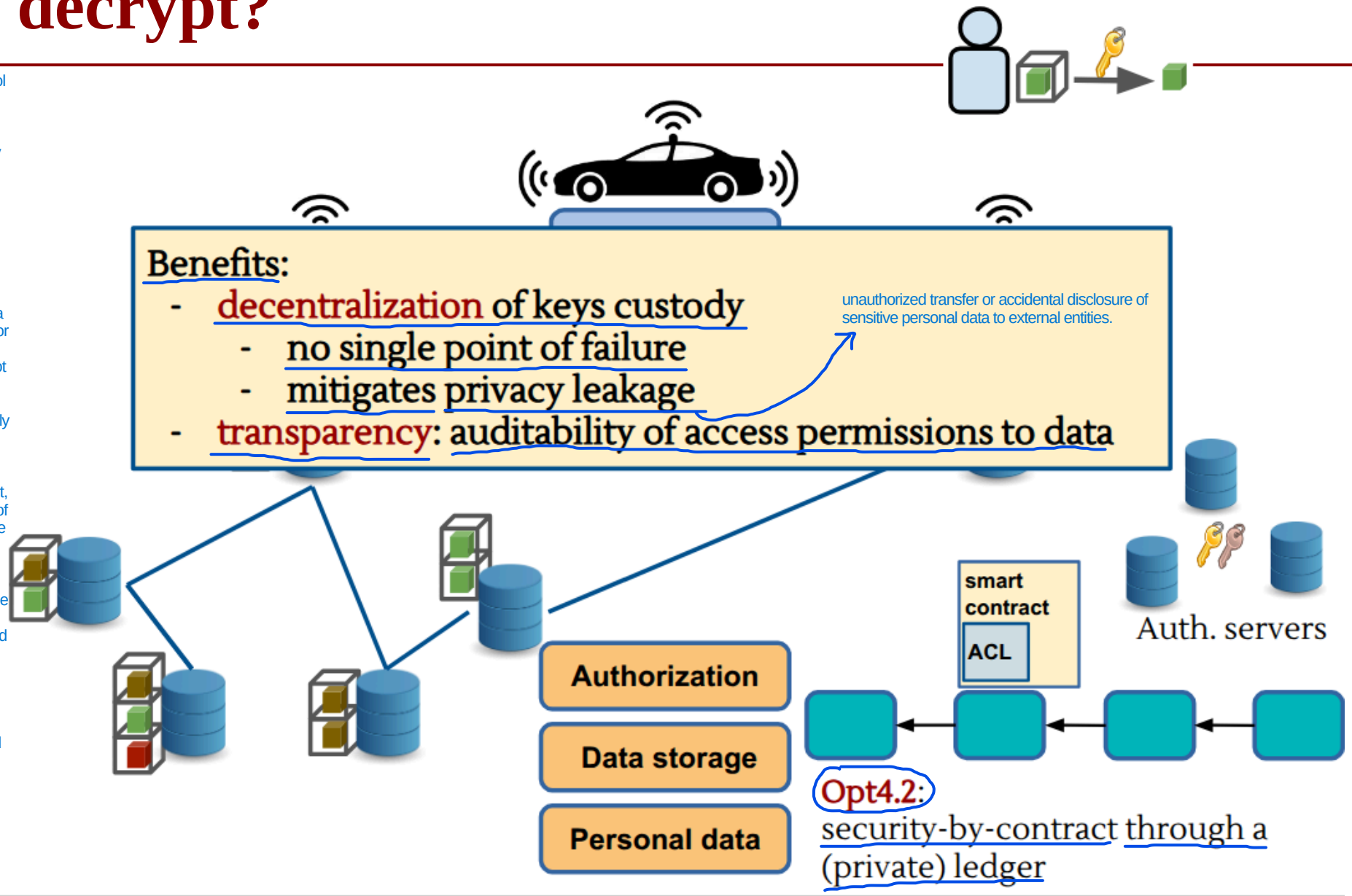


How to decrypt?

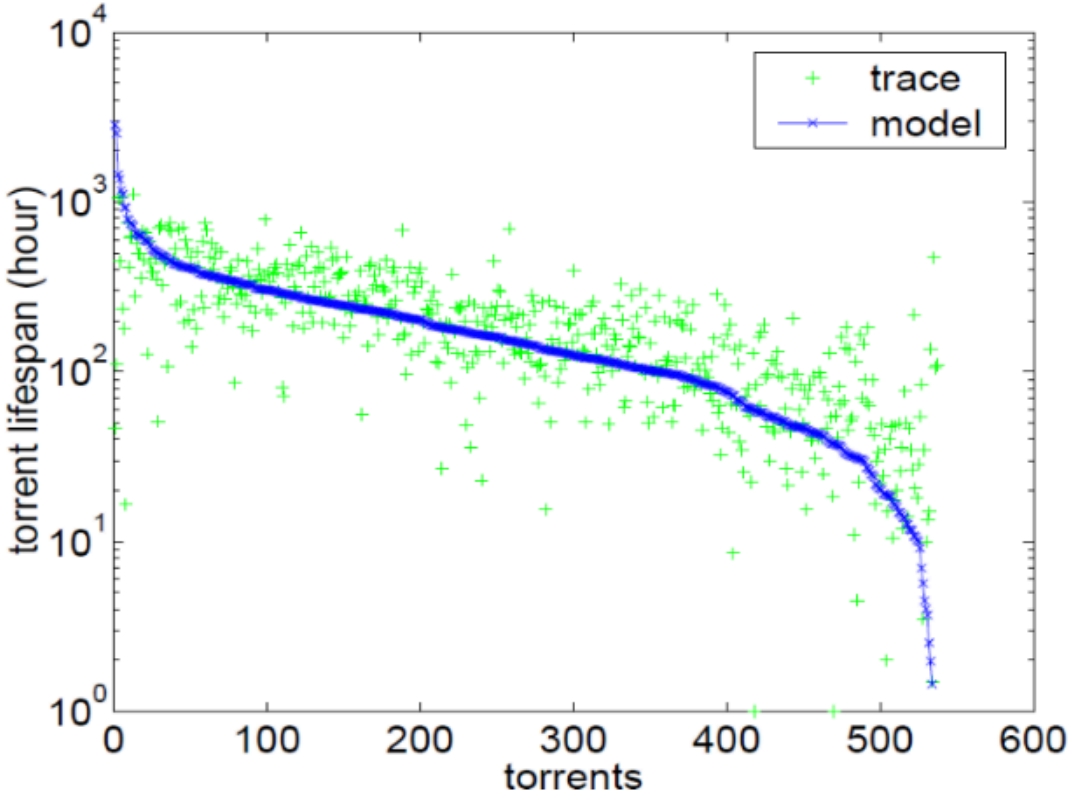
Opt4.2 solves the problem of access control in a decentralized environment. Instead of entrusting the decision of who can view the data to a traditional central server (which represents a single point of failure), security rules are written as code within a smart contract that resides on a private ledger (a private blockchain managed by trusted entities such as municipalities and authorities).

The Workflow
1. The Request: An application wants to access a sensitive mobility file and makes a request via a network authentication node or server.
2. Contract Verification: The server does not decide on its own, but queries the Smart Contract on the Private Ledger. The Smart Contract evaluates the request automatically and incorruptibly, verifying whether the requester complies with the established digital agreements.
3. Key Release: If the requirements are met, the Smart Contract authorizes the release of the cryptographic key needed to decrypt the file.
4. Download and Decryption: The application downloads the encrypted file from the DFA using its Content Address (the Hash) and, thanks to the key obtained via the Smart Contract, can finally decrypt it and read it in plain text.

Why is it used? (Benefits for the exam)
- Confidentiality: No unauthorized user can decrypt the raw data.
- No Single Point of Trust: There is no need to trust a single system administrator; the Smart Contract's mathematical code enforces the rules transparently.
- Auditability: Every time access to data is requested, the action invokes the Smart Contract and leaves an immutable transaction on the Ledger. In the event of abuse, there is a perfect historical record of who accessed what and why.



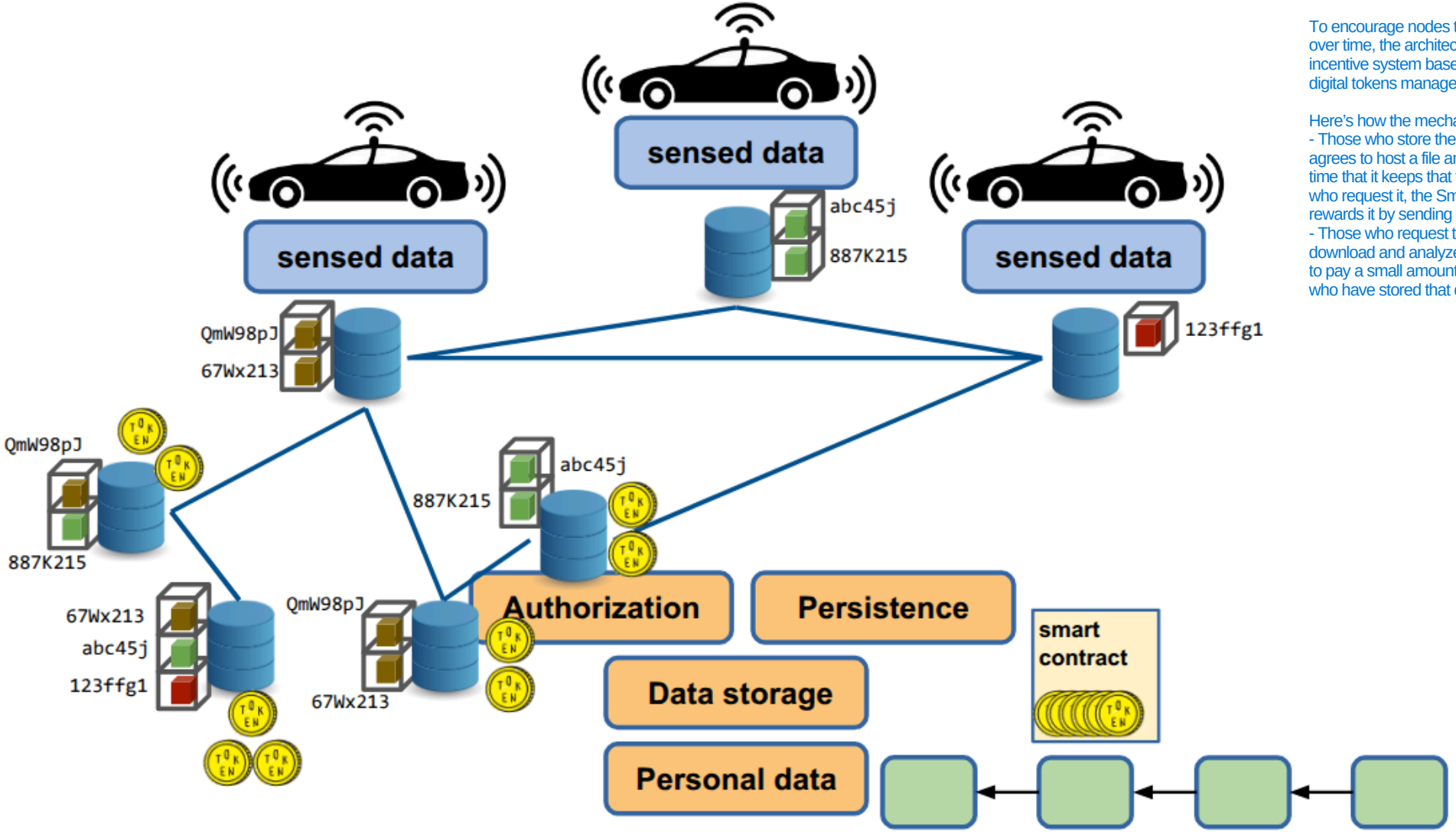
Data persistence: we'd like to avoid this



Guo, Lei & Chen, Songqing & Xiao, Zhen & Tan, Enhua & Ding, Xiaoning & Zhang, Xiaodong. (2007). A performance study of BitTorrent-like peer-to-peer systems. Selected Areas in Communications, IEEE Journal on. 25. 155 - 169. 10.1109/JSAC.2007.070116.

Data persistence

Incentives to cooperate:
blockchain based tokens
E.g. Filecoin, Sia, Storj, Swarm (??)



To encourage nodes to "cooperate" and maintain data over time, the architecture introduces an economic incentive system based on tokens (cryptocurrencies or digital tokens managed by the blockchain).

Here's how the mechanism works:
- Those who store the data earn: if a network node agrees to host a file and mathematically proves over time that it keeps that file intact and accessible to those who request it, the Smart Contract automatically rewards it by sending tokens.
- Those who request the data pay: anyone wishing to download and analyze historical mobility data will have to pay a small amount of tokens to compensate those who have stored that data.

Threat Analysis

Integrity Attacks

- Injection of false mobility data
- Hash substitution attempts
- Replay attacks

Mitigation

- Digital signatures at source
- Hash anchoring on DLT
- Verifiable timestamps

Threat Analysis

Confidentiality Attacks

- Re-identification via mobility traces
- Linkability of pseudonymous identifiers
- Key leakage

Re-identification via mobility traces: Even if GPS data is stripped of the user's first and last name, travel patterns reveal unique habits. If an attacker notices a trace that departs every morning at 8:00 AM from a specific residence and arrives at 8:30 AM at a specific office, by cross-referencing this data with the phone book or social networks, they can accurately trace the driver's real identity.

Linkability of pseudonymous identifiers: If a user consistently sends data using the same pseudonym (e.g., a fixed ID or a constant public key such as User_99), a malicious actor can "connect the dots" (linkability). Even without knowing the user's real identity, the attacker can reconstruct the entire history of all their past movements over the course of months, creating a perfect profile of them.

Key leakage: If a user mishandles their private key on their phone, or if the Smart Contract's key management system (Opt4.2) is compromised, an attacker who gains access to the keys can instantly decrypt all historical files stored on the DFS (Opt4), rendering all defenses useless.

Mitigation

- Strong encryption
- Key rotation
- Minimal use of metadata

Strong encryption: Raw mobility data is encrypted directly on the user's device using robust, standardized algorithms before being transmitted. This way, if a node in the distributed storage system is compromised, the attacker will find only incomprehensible bits, preventing key leakage to third parties.

Key rotation: To break linkability, the system never uses the same key or identifier indefinitely. Encryption keys and user pseudonyms are changed (rotated) continuously and automatically (e.g., every hour or at the start of each new trip). If an attacker manages to compromise a key, they will only be able to decrypt a very small fragment of a route lasting a few minutes, without being able to link it to the same user's past or future routes.

Minimal use of metadata: The principle of data minimization is applied. When the file is prepared and the hash is sent to the Ledger, all ancillary data not strictly necessary for calculating traffic is deleted (e.g., the car model, the phone's operating system, or the device's constant ID). The less metadata there is, the harder it is for an attacker to carry out correlation and re-identification attacks.

Threat Analysis

Availability Attacks

- Denial-of-Service against DFS nodes
- Ledger congestion
- Incentive collapse

Denial-of-Service against DFS nodes (DoS against DFS nodes): An attacker floods the Decentralized File System (DFS) nodes (Opt4) that host mobility files with network requests. If these nodes become overloaded, they can no longer distribute files to users or the provider, rendering the entire transport monitoring service unusable.

Ledger congestion: Every time a vehicle generates a file, it must send the hash to the Ledger to “anchor” it (Opt3). During peak hours (e.g., at 8 a.m.), millions of cars send transactions simultaneously. If the Ledger (even if private) becomes congested, the wait times for transaction confirmation increase dramatically (high latency). So, Data is not recorded in a timely manner, making it impossible to monitor traffic in real time.

Incentive collapse: This is linked to Data Persistence. If the economic value of tokens collapses, or if the software reward mechanism fails, the nodes in the P2P network no longer have any economic incentive to waste space and bandwidth to store files. The nodes shut down their servers, and historical data disappears.

Mitigation

- Replication policies
- Multi-node anchoring
- Layered fallback mechanisms

Replication policies: To counter DoS attacks on DFS nodes, the system enforces redundancy rules. Whenever a user uploads a file, it is not saved on a single node, but is automatically copied and replicated across dozens of geographically dispersed nodes. If an attacker takes a node offline, the network will instantly retrieve the same file (uniquely identified by its Content Address) from any of the other active mirror nodes.

Multi-node anchoring: Instead of requiring all vehicles to send their hashes to a single ledger server or gateway, the in-vehicle application is designed to interface with a dynamic pool of validator nodes on the ledger.

Layered fallback mechanisms: If the main decentralized network experiences total congestion or an incentive collapse, the system seamlessly scales to backup layers. For example, if the Ledger is temporarily blocked, the user’s device locally buffers the hashes in a queue (local buffering) and then sends them in bulk as soon as the network clears, preventing the immediate loss of mobility flows.

Lessons Learned (Crowdsensing)

What really matters

- Data is **distributed**, but risks are **centralized through aggregation**
- **Integrity, privacy, and availability** must be designed together
- Architecture choices directly impact **trust assumptions**

The key concept: Every architectural choice shifts the “trust boundary.” If you choose a traditional centralized solution, your assumption is: “I blindly trust the central server.” If you choose Opt4.2, your assumption changes: “I don’t trust anyone; I only trust the mathematics of the Ledger and the Smart Contract.” The architecture defines who (or what) you must trust.

What we need to study

- Secure data sharing
- Distributed systems
- Privacy-preserving techniques (anonymization, aggregation)

Use Case #2: Security Analysis of a Clinical ML System

Scenario

Federated Learning:
- Data remains local: Hospital A, Hospital B, and Hospital C keep their medical records and intensive care unit sensor data on their own secure servers.
- Distributed training: A central cloud orchestrator sends a base version of the diagnostic model to each hospital. Each hospital trains the model locally using only its own data.
- Parameter aggregation: Hospitals do not send patient data to the cloud, but only the mathematical weights. The cloud orchestrator aggregates these parameters to create an improved global model, which is then redistributed to everyone (the weights).
- Clinicians then query this global model via a Web API: they send the current data of a patient in the ICU, and the API instantly returns the percentage risk of readmission.

A diagnostic model is trained on data collected from **multiple hospitals**

The model is deployed as a **cloud-orchestrated federated learning system** and accessed through a web API by clinicians

Function: Predict ICU (Intensive Care Unit) readmission risk

Predicting Intensive Care Unit (ICU) readmission risk refers to using data analytics and machine learning models to calculate the probability that a patient, once discharged from the ICU to a standard hospital unit or home, will deteriorate and require readmission to the ICU within a specific timeframe (typically 48 to 72 hours, or up to 30 days).

The system must satisfy:

- Regulatory constraints (GDPR compliance)

- Data Minimization & Sovereignty: Raw patient data never leaves the physical and legal boundaries of the originating hospital.
- Privacy-Preserving Techniques: Since a sophisticated attacker could theoretically "reconstruct" certain patient data by analyzing the mathematical weights sent to the cloud, techniques such as Differential Privacy (adding controlled statistical noise to the gradients) or Homomorphic Encryption (aggregating parameters in encrypted form in the cloud) are used.

- Clinical reliability requirements

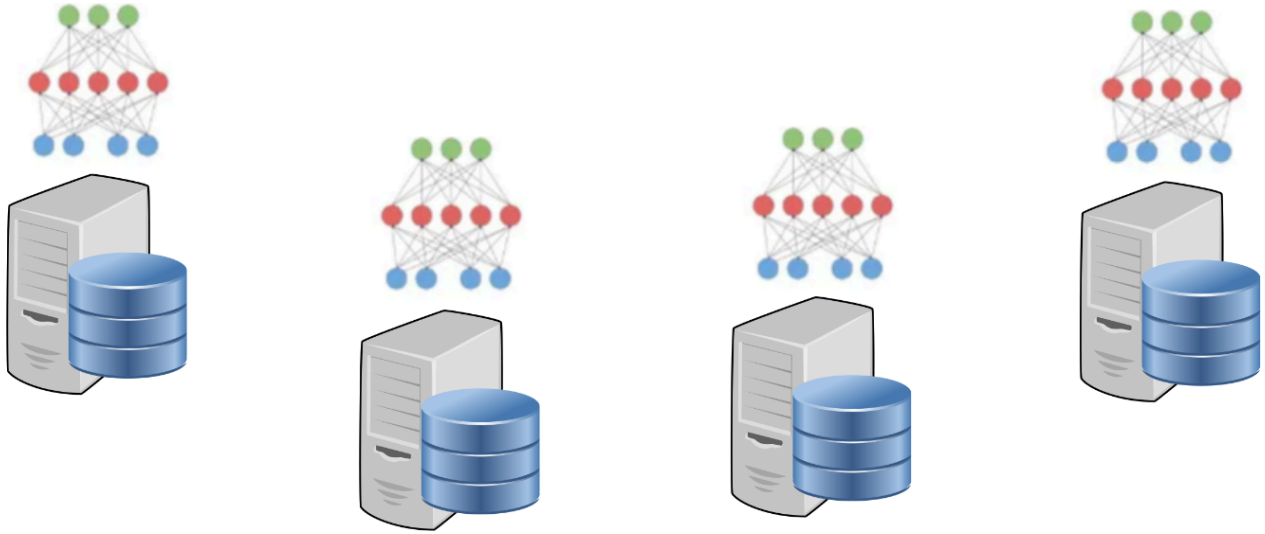
- Domain Bias Mitigation: Hospitals have different patient populations, equipment, and clinical guidelines. The federated model must be validated to prevent metrics from a large hospital from overshadowing the specific characteristics of a smaller clinic.
- Explainable AI (XAI): The API should not only provide a percentage (e.g., "Re-admission risk: 85%"), but should also explain which clinical parameters (blood pressure, ejection fraction, blood parameters) drove the prediction, allowing the physician to verify the clinical rationale behind the alert.

- Security and resilience against adversarial manipulation

System Architecture

Training Paradigm: **Federated Learning** (FL) across N hospitals

of the model



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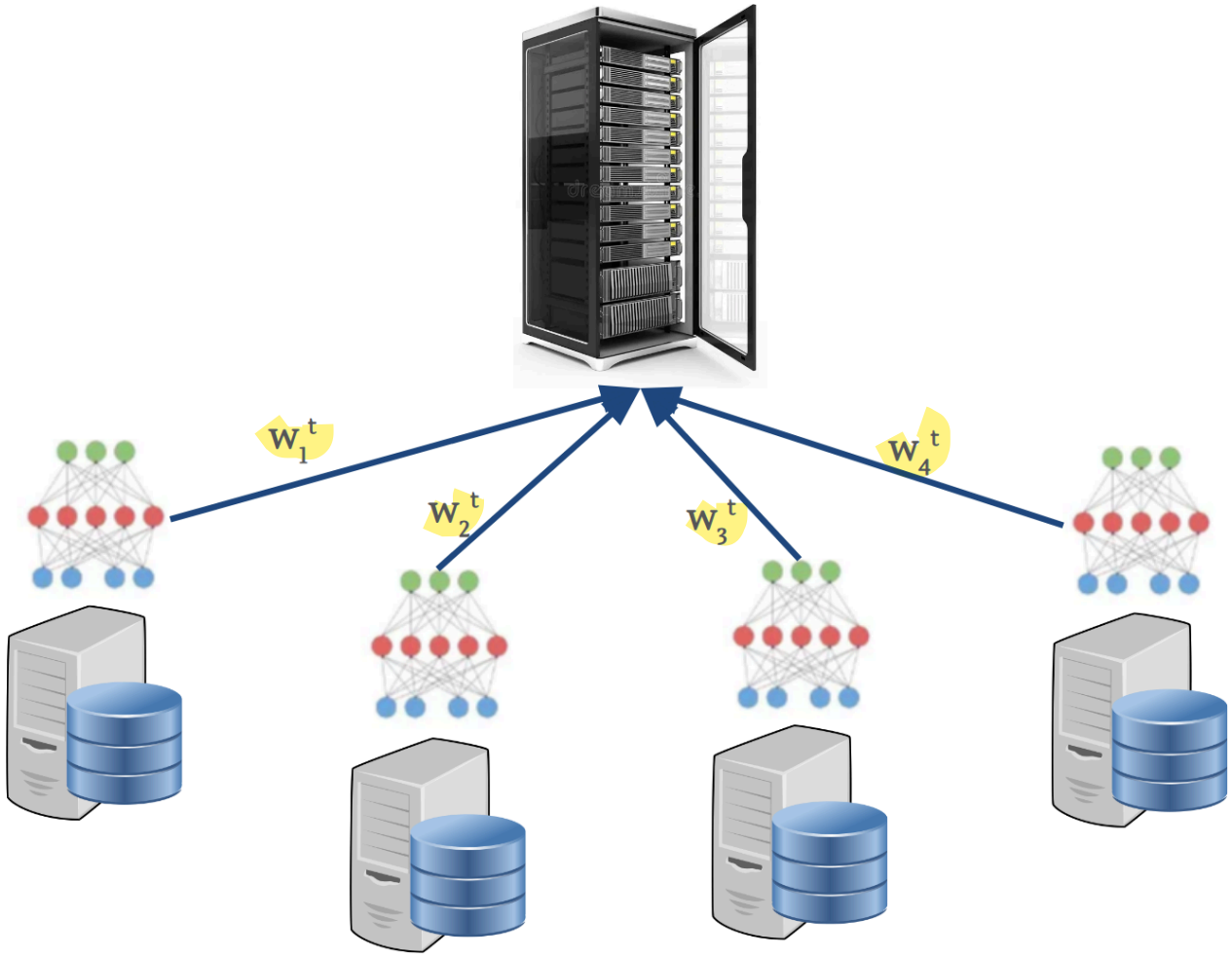
UNIVERSITÀ DI BOLOGNA

System Architecture

Training Paradigm: **Federated Learning (FL)** across N hospitals



System Architecture

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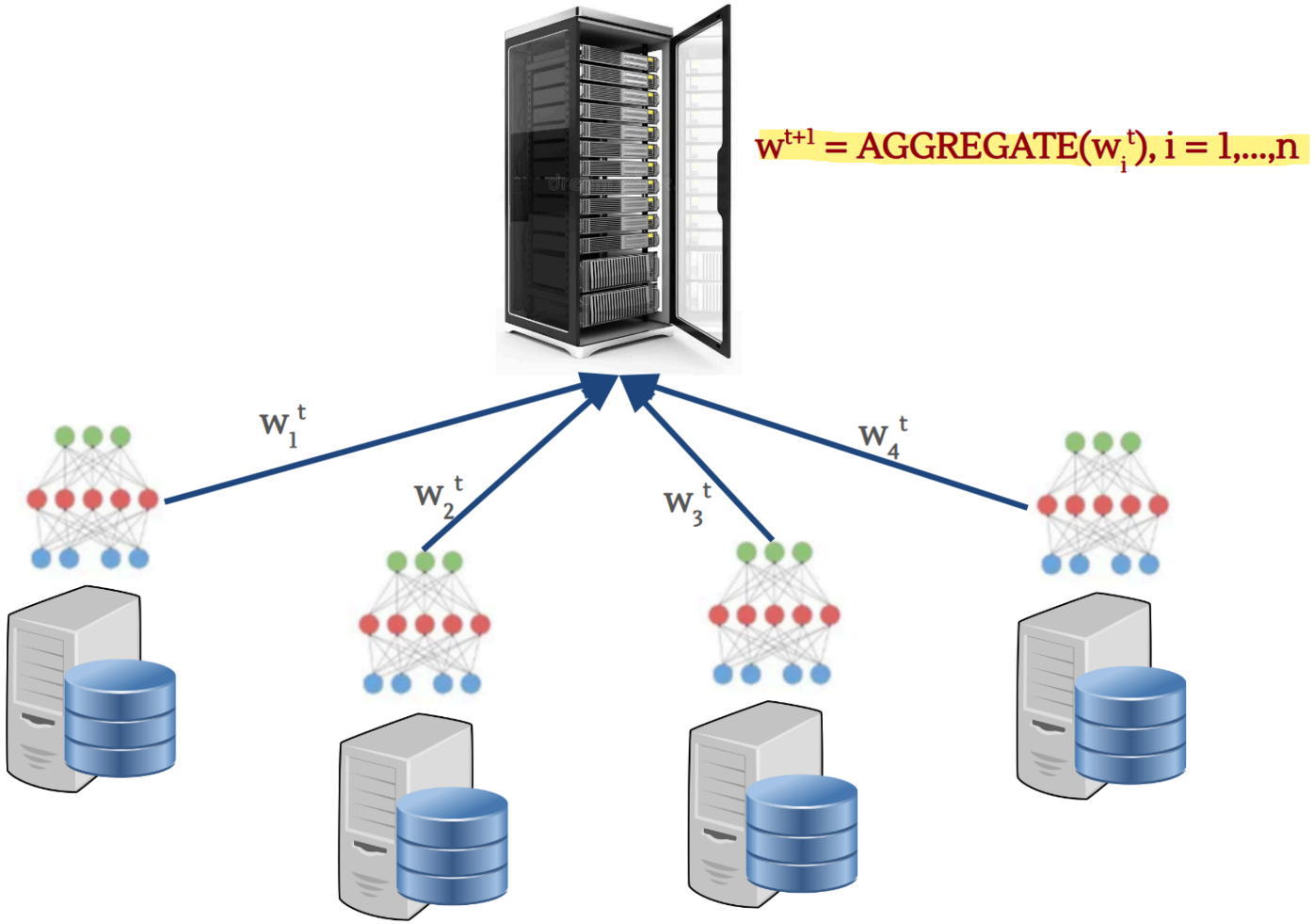

System Architecture

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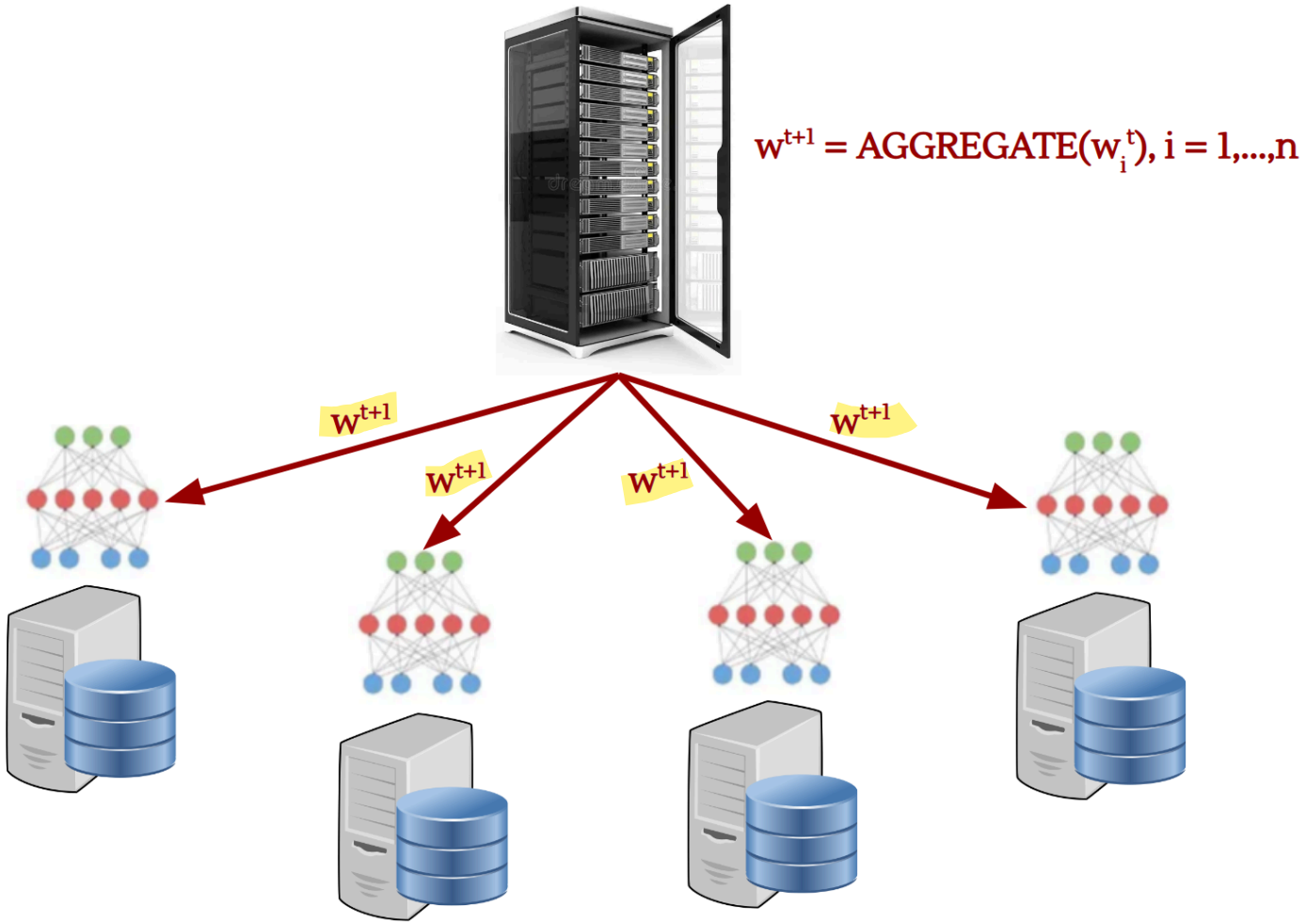


System Architecture

Data Characteristics

- Patient records remain local to each hospital
- Model updates (gradients or weights) are transmitted to a central aggregator
- During **training**, labels are produced by external annotation contractors that have access to local data

Deployment

- Cloud-based infrastructure
- API-based access for clinical users

Threat Modeling Framework

We classify threats according to three primary security objectives:

- **Integrity**: correctness of model parameters and outputs
- **Privacy**: protection of patient information
- **Availability**: continuity and reliability of clinical service

Integrity Threat 1: Data Poisoning

Attack Surface

Training labels provided by external contractors

Integrity Threat 1: Data Poisoning

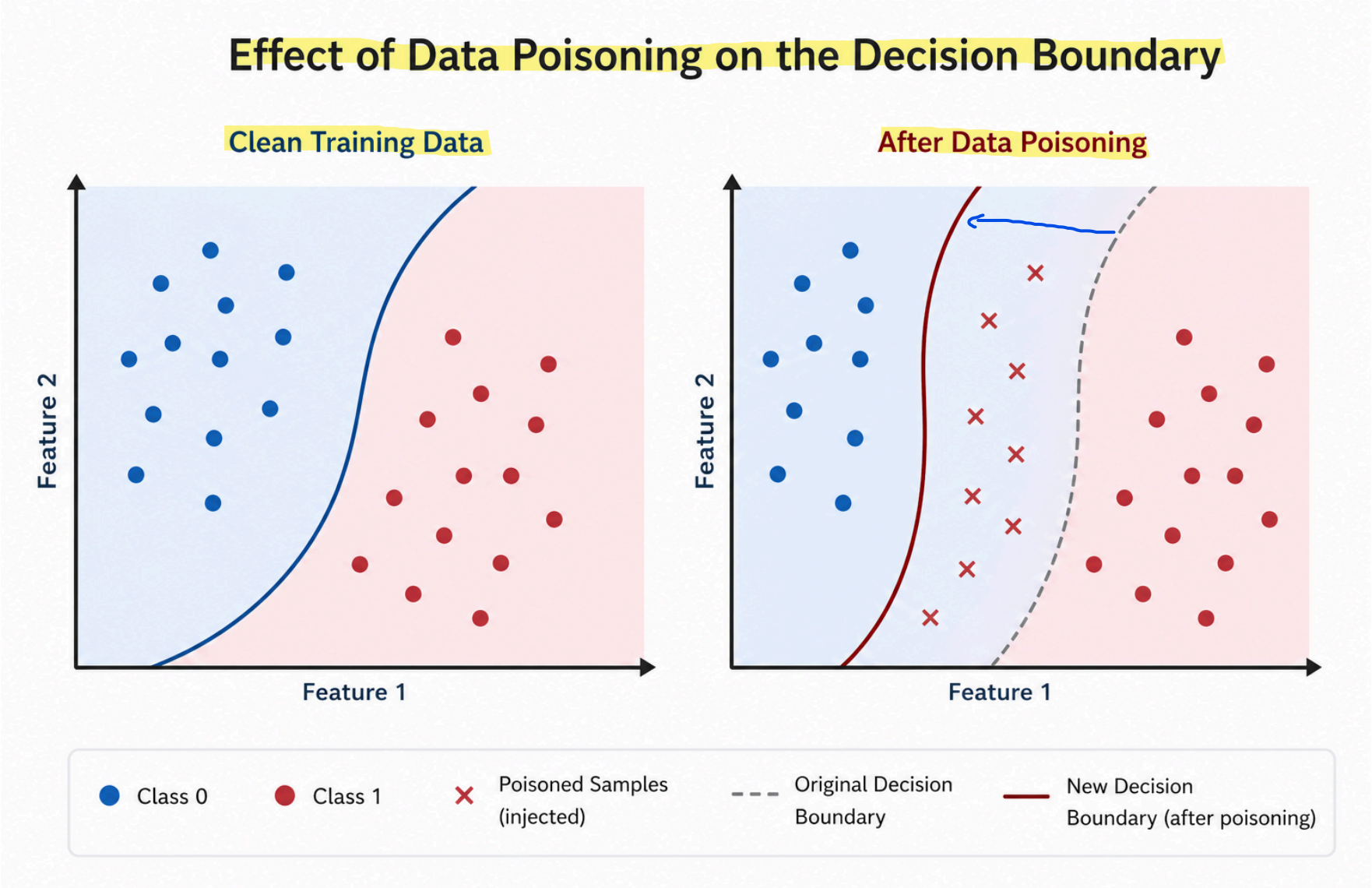
Attack Surface

Training labels provided by external contractors

Threat Description

- Incorrect or maliciously altered labels degrade model correctness
- Systematic bias may be introduced in specific sub-populations

Integrity Threat 1: Data Poisoning



Integrity Threat 1: Data Poisoning

Measurable Mitigation

Valutazione dell'affidabilità tra valutatori

Inter-Rater Reliability (IRR) Scoring

- Each record labeled by ≥ 3 annotators
- Discard or re-label samples below a defined agreement threshold

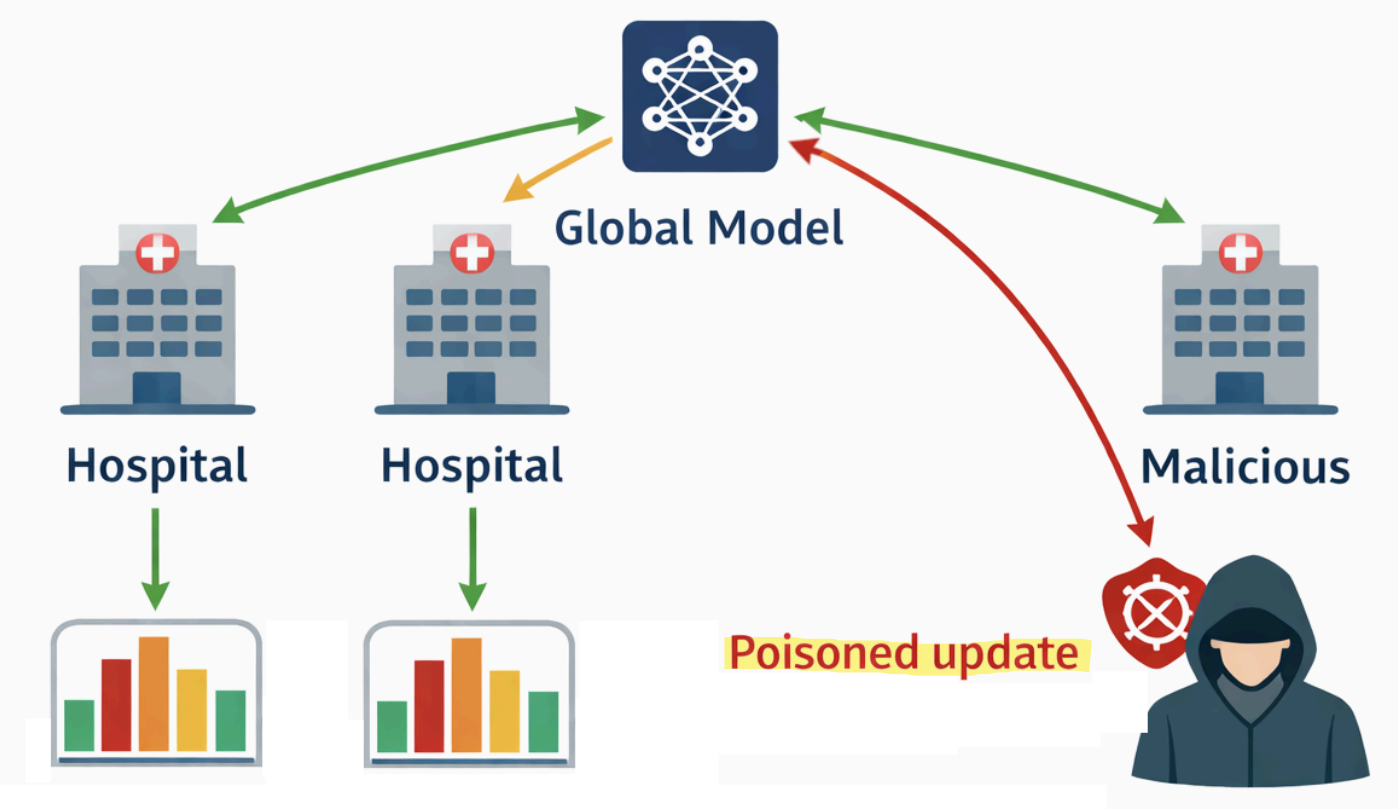
Trade-off

- Increased annotation cost (approximately linear in redundancy factor)
- Increased dataset preparation latency
- Possible reduction in effective dataset size

Integrity Threat 2: Poisoned Federated Updates

Attack Surface

Model updates transmitted by hospitals in the FL protocol



Integrity Threat 2: Poisoned Federated Updates

Attack Surface

Model updates transmitted by hospitals in the FL protocol

Threat Description

A compromised participant sends adversarial gradients to bias the global model

Effects may include:

- Targeted backdoor behavior → How it works: The attacker secretly "teaches" the model to behave abnormally only in the presence of a specific trigger (backdoor), while in all other cases the model continues to function normally.
  Clinical example: The overall model correctly predicts 85% of readmission risks, but the attacker has inserted a hidden rule: if the patient is exactly 64 years old and has a specific hemoglobin value, the model will always predict "Low Risk," even if the patient is in critical condition. This allows an attacker to intentionally manipulate the diagnosis of a specific target without being detected by general accuracy checks.
- Systematic degradation of specific predictions → How it works: The attack aims to undermine the model's reliability across entire categories of data by introducing a strong bias (distortion) or drastically reducing its accuracy for specific groups.
  Clinical example: Poisoned gradients destabilize training related to patients with heart disease. As a result, the model becomes unable to correctly predict readmission to the ICU for heart patients, leading to erroneous medical decisions on a systemic scale for that specific category of patients.

Integrity Threat 2: Poisoned Federated Updates

Measurable Mitigation

Robust Aggregation Rules

In standard federated learning, the classic algorithm (FedAvg) calculates a simple weighted average of all the gradients received. If a hospital sends a “poisoned” gradient with extremely large or distorted values, the average is heavily distorted. Robust Aggregation Rules replace the classic average with advanced mathematical functions capable of withstanding Byzantine attacks (attackers)

Examples:

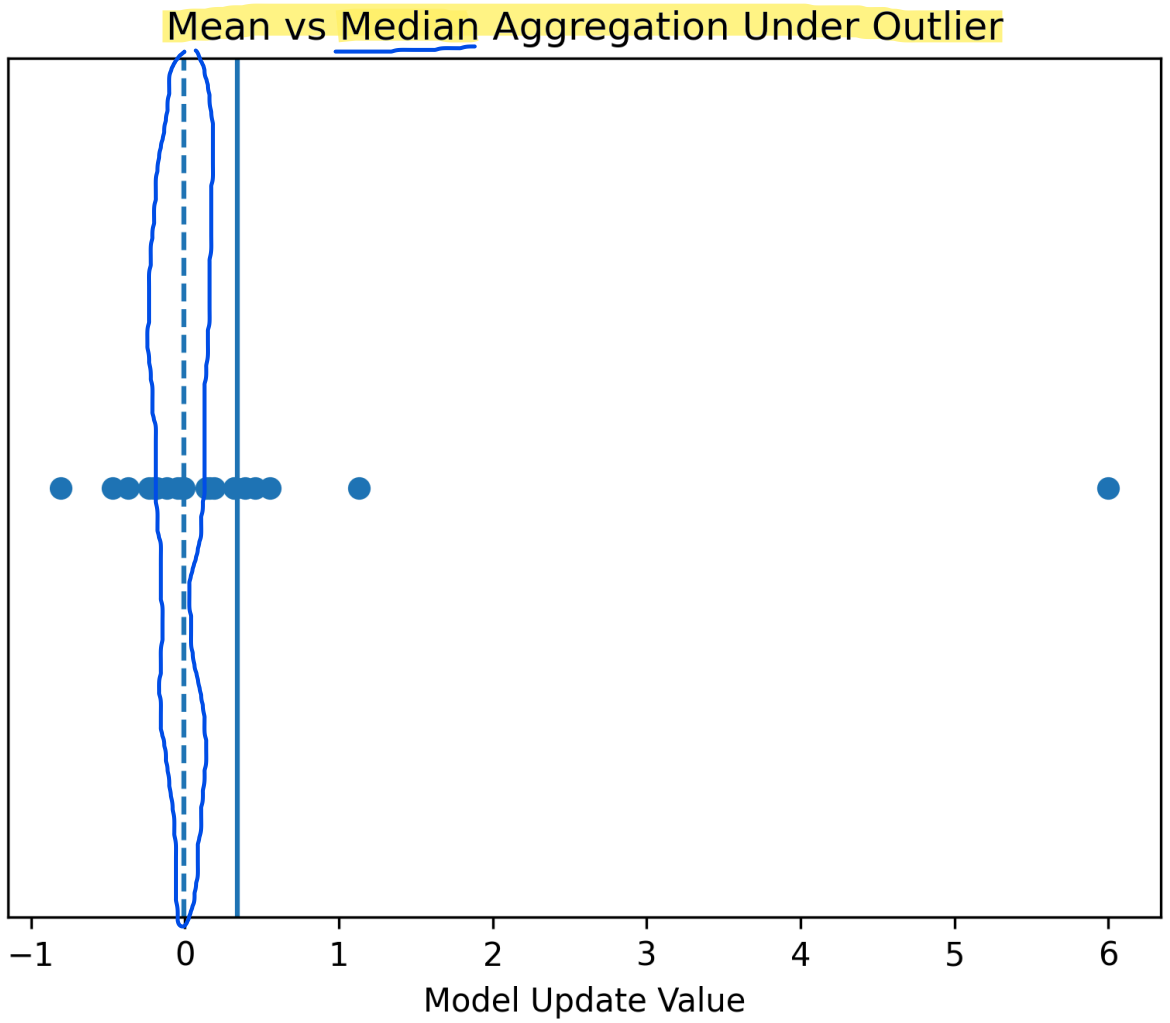
- Krum
- Coordinate-wise median
- Trimmed mean

Metric:

- Bounded influence of outlier updates

This is the fundamental metric for measuring the effectiveness of the defense. A system is considered secure if the maximum impact that a malicious node can have on the global model is strictly limited and bounded. By applying Krum, the Median, or the Trimmed Mean, the mathematics ensures that even if an attacker sends gradients with infinitely large or heavily distorted values to create a backdoor, the aggregation algorithm will nullify or reduce their influence to a negligible amount, preserving the integrity and clinical reliability of the final diagnostic model.

Integrity Threat 2: Poisoned Federated Updates



Integrity Threat 2: Poisoned Federated Updates

Trade-off

- Reduced sensitivity to legitimate distributional heterogeneity
- Potential degradation of performance on minority subpopulations
- Increased aggregation complexity

Privacy Threat: Membership Inference

Attack Surface

Public-facing inference API

Threat Description

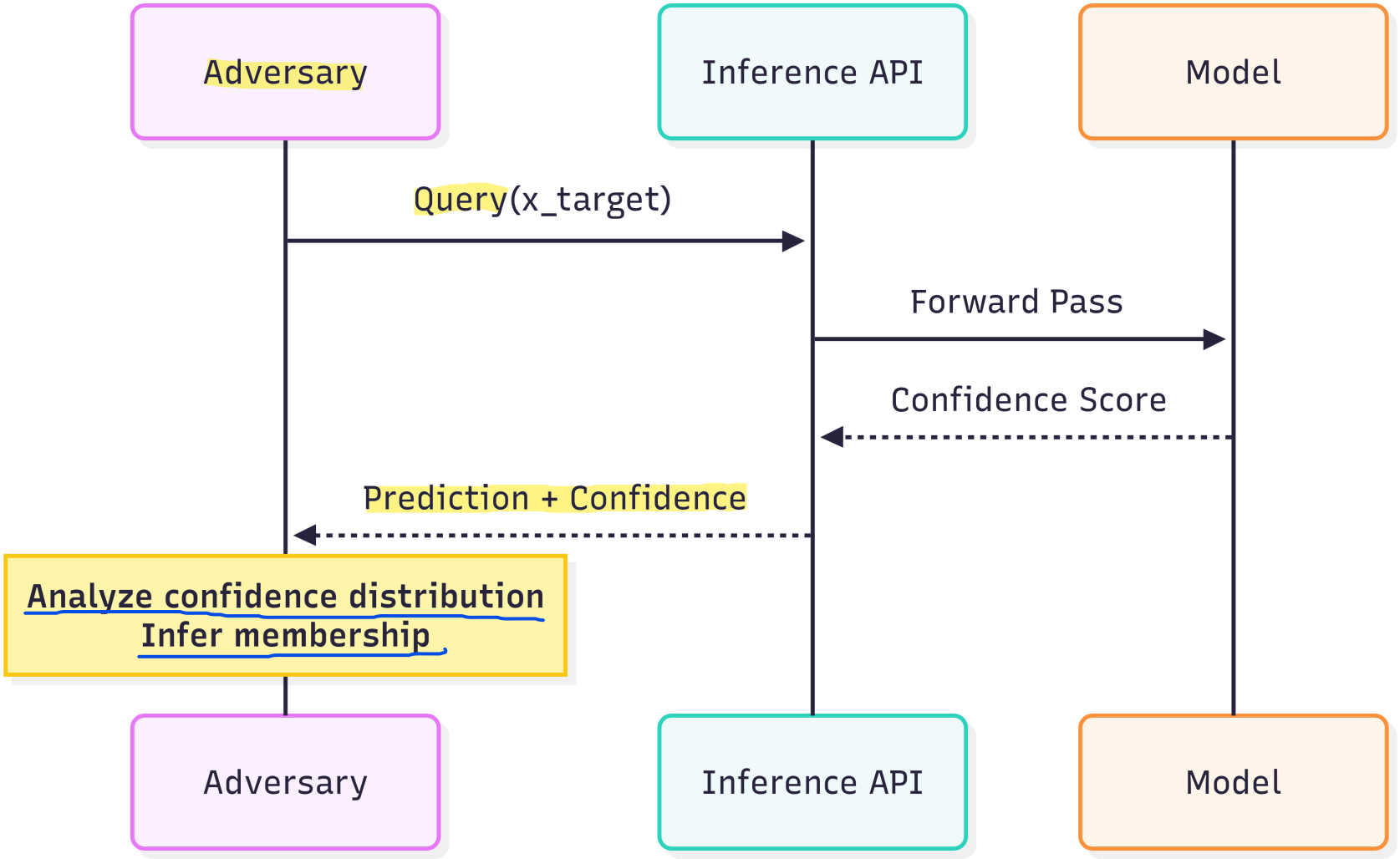
An adversary queries the model to determine:

- Whether a specific patient record was used in training
- if it is possibile that ~~Whether~~ sensitive attributes can be reconstructed

Relevant attacks:

- Membership inference
- Model inversion

Privacy Threat: Membership Inference



Privacy Mitigation: Differential Privacy

Mechanism

The fundamental principle is based on the concept of indistinguishability: Imagine you have a dataset D (e.g., the clinical data of patients at a hospital) and a neighboring dataset D', which contains exactly the same data as D, except for a single record (for example, it adds or removes the data of a single patient). An AI model satisfies Differential Privacy if, when run on D or on D', it produces an output with an almost identical probability distribution. In simple terms: If the result of the medical analysis or the model does not change significantly whether I participate in the study or not, an attacker observing the output will never be able to tell whether my personal data was present in the system or not, neutralizing the risks of re-identification.

- **Differential privacy (DP)** is a *mathematically rigorous privacy definition* where one dataset vs its neighbor should not significantly change the output probability distribution
- DF is commonly described via the (ε, δ)-condition

- Epsilon: Controls the level of privacy. It represents the maximum multiplier of the probability difference between the two datasets. The smaller epsilon is (closer to zero), the more identical the output across the two datasets will be, ensuring very strong privacy. On the other hand, an epsilon that is too small introduces a lot of noise, reducing the model's clinical accuracy.
- Delta: Represents the probability that the strict privacy constraint will fail (a sort of "margin of error" or probability of a small information leak). In secure systems, delta must be extremely small.

Formal Guarantee

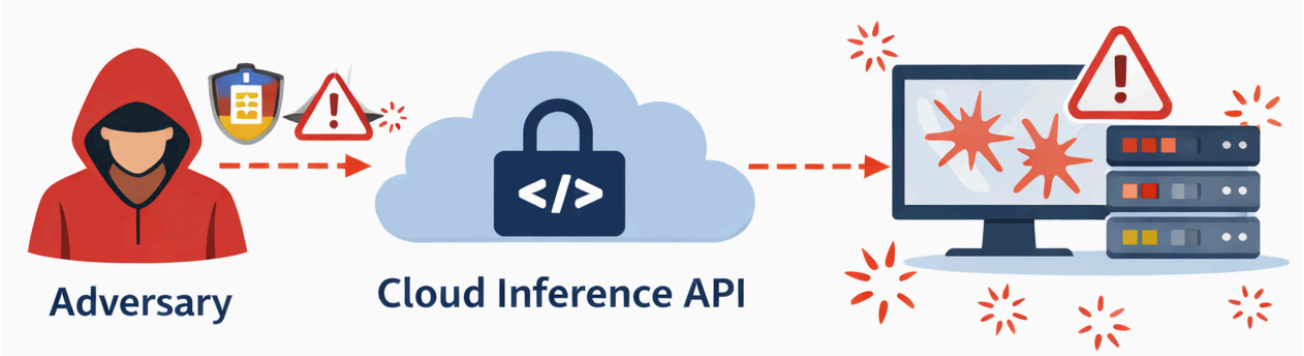
For adjacent datasets D, D' differing in one record:

$$P(M(D) \in S) \leq e^{\varepsilon} P(M(D') \in S) + \delta$$

Availability Threat: Adversarial Denial of Service

Attack Surface

Cloud inference API



Threat Description

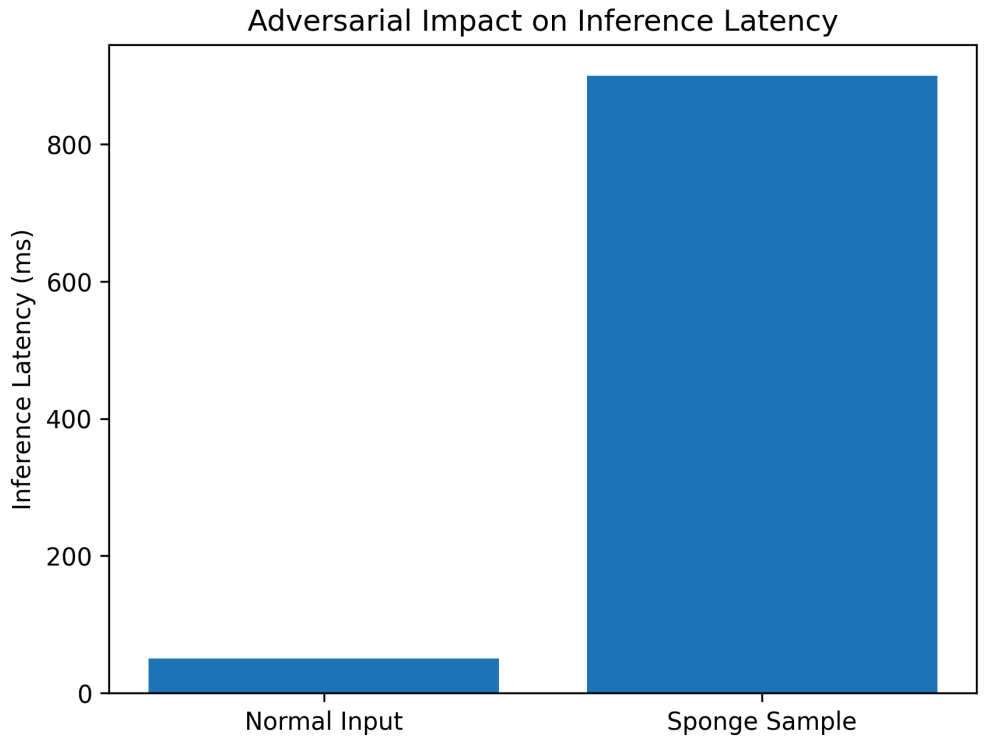
Adversarial inputs crafted to:

- Maximize computational load ("sponge samples")
- Trigger numerical instability
- Cause repeated exceptions

Availability Threat: Adversarial Denial of Service

Effect

- Increased latency
- Degraded clinical workflow refers to a breakdown in standard patient care processes
- Potential service outage



Controls

1. API rate limiting per authenticated user
2. Input validation and anomaly detection
3. Resource caps per inference request

Availability Mitigation

Measurable Metrics

- Maximum latency per request
- Rejection rate of anomalous inputs
- Service uptime percentage

Trade-off

- False rejection of legitimate but atypical clinical cases
- Operational friction in emergency contexts

Lessons Learned (Clinical ML)

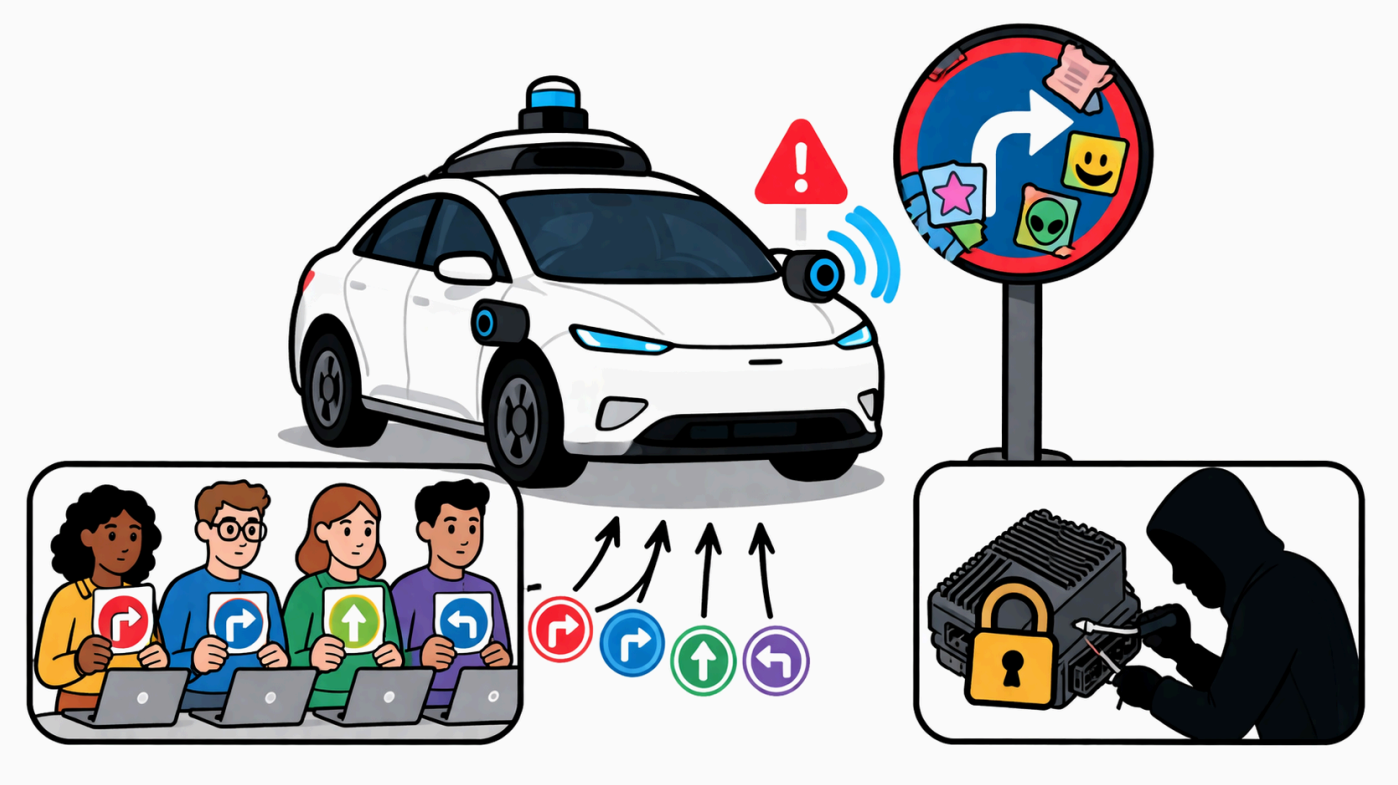
What really matters

- ML systems introduce **new attack surfaces** (training + inference)
- Security is **data-dependent**, not only system-dependent
- Privacy risks persist even without direct data access

What we need to study

- Trustworthy AI (adversarial ML, poisoning, evasion, inference attacks)
- Differential Privacy
- Federated Learning security
- Evaluation under adversarial settings

Use Case #4: Security Analysis of an Autonomous Perception System



Scenario

A perception model for an **autonomous vehicle** is deployed on an embedded ECU (Electronic Control Unit) with:

- 4 GB RAM
- Limited compute budget
- Infrequent software update capability

The system must remain reliable under:

- Sensor noise and weather variability
- Physical-world adversarial manipulation (e.g., adversarial stickers)
- Supply-chain risks in the annotation pipeline

System Architecture

Function: Traffic sign recognition and scene perception

Deployment Paradigm: Edge inference + selective cloud interaction

System Architecture

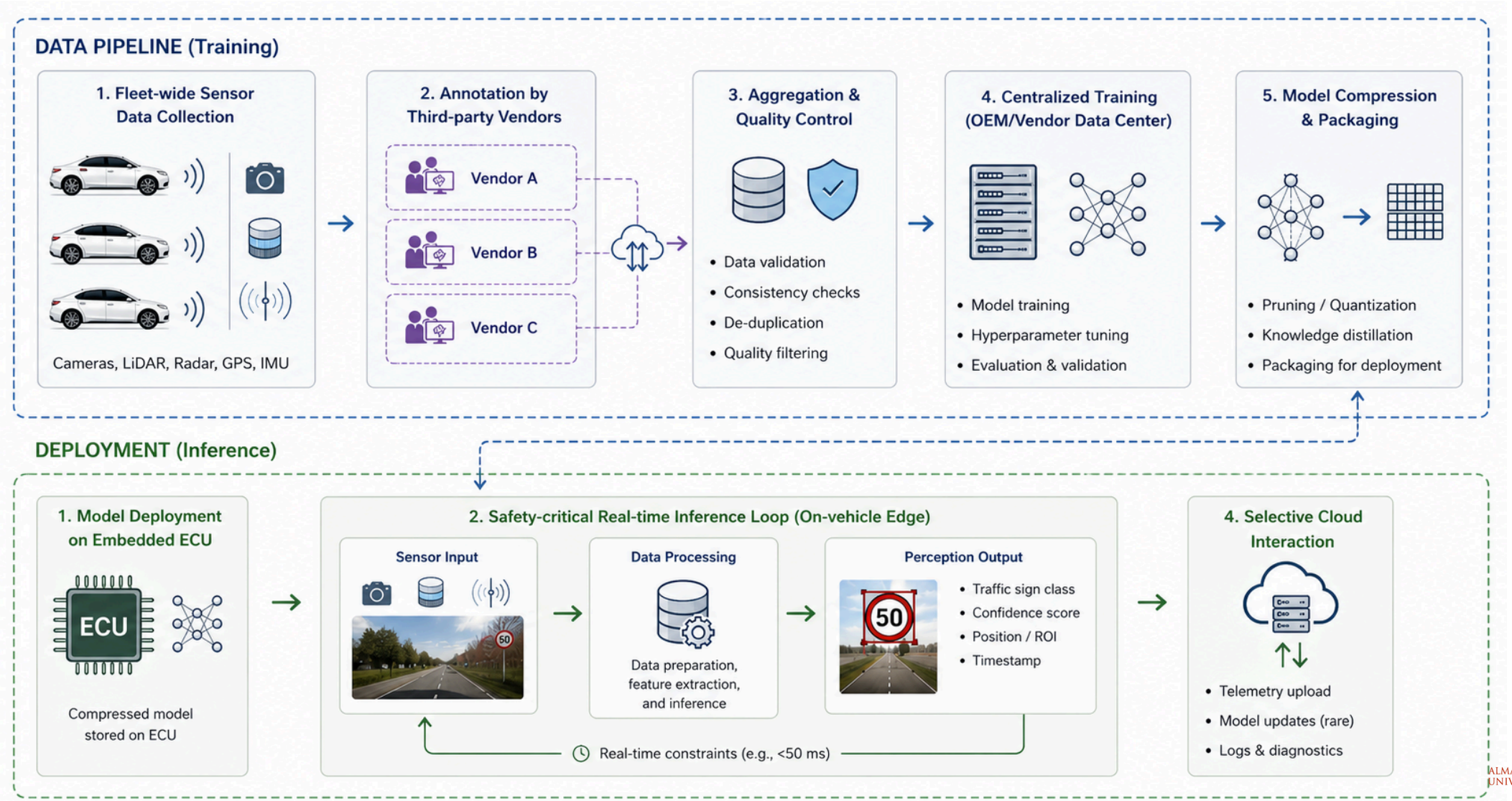
Data Pipeline

- Fleet-wide sensor data collection
- Annotation performed by three third-party vendors
- Centralized training performed in the vendor data center

Deployment

- Compressed model deployed on embedded ECU
- **No** continuous online retraining
- Safety-critical real-time inference loop

System Architecture



Threat Modeling Framework

Security objectives:

- **Integrity**: Correctness of model parameters and outputs
- **Availability**: Real-time inference under adversarial conditions
- **Authenticity**: Assurance that deployed model is untampered

Integrity Threat 1

Physical Adversarial Patches

Attack Surface

Road signs modified with localized adversarial stickers

Integrity Threat 1

Threat Model

An adversary applies a perturbation δ such that:

$$\| \delta \|_\infty \leq \epsilon$$

but:

$$f(x) \neq f(x + \delta)$$

Note: $\| x \|_\infty = \max | x_i |$

Robustness-Capacity Trade-off

Certified robustness typically requires:

- Robustness to $\| \delta \|_\infty \leq \epsilon \rightarrow$ deep neural nets

  To enforce stability over perturbation regions
- Deep NNs $\rightarrow$ longer training time

  Robust training involves additional constraints and computations
- Overfitting avoidance $\rightarrow$ reduced accuracy on clean data

  Smoother decision boundaries reduce sensitivity but also precision

Integrity Threat 2

Annotation Supply-Chain Poisoning

Attack Surface

Labels provided by three external vendors

Threat Description

- Systematic label corruption
- Targeted backdoor triggers
- Class-conditional bias injection

Potential consequences

- Misclassification of specific traffic signs
- Hidden backdoor activation

Integrity Threat 2: Mitigation

Controls

1. **Redundant Annotation**
    - Each sample labeled by ≥ 3 annotators
2. **Agreement Scoring**
    - Multi-rater consistency
    - Reject samples with consistent labels below threshold
3. **Vendor Drift Monitoring**
    - Per-vendor confusion matrix tracking
    - Statistical deviation detection

Integrity Threat 2

Trade-off

- Increased annotation cost
- Dataset size reduction
- Increased preparation latency

Availability Threat

Adversarial Sensor Inputs

Attack Surface

Camera and LiDAR streams

Threat Description

Inputs crafted to:

- Trigger worst-case compute paths
- Induce latency spikes
- Cause frame drops beyond control deadlines

Consequence:

- Missed real-time constraints
- Unsafe control transitions

Availability: Mitigation

Controls

1. Input validation and preprocessing constraints
2. Worst-case latency profiling
3. Resource isolation at process level

Metrics

- Maximum inference latency (ms)
- 99.9th percentile latency under adversarial load
- Frame drop rate

Trade-off:

- False rejection of legitimate rare scenarios

Authenticity Threat

Model Tampering on ECU

Attack Surface

Model binary stored in on-device flash memory

Threat Description

- Unauthorized firmware modification
- Parameter replacement
- Backdoored model injection

Authenticity Mitigation: Trusted Execution

Controls

1. **Secure Boot**
    - SHA-256 integrity check of firmware at load time
2. **Trusted Execution Environment (TEE)**
    - Isolated inference runtime
    - Remote attestation

Metrics

- Integrity verification

Trade-off

- Increased startup latency
- Hardware cost overhead

Lessons Learned (Autonomous Systems)

What really matters

- The physical world becomes an **attack surface**
- Constraints (latency, memory) limit defenses
- Supply chain is part of the threat model

What we need to study

- Robustness of ML models
- Secure deployment (TEE, secure boot)
- Real-time constraints and worst-case analysis
- Data pipeline security

Use Case #5: Overlay Networks of Interacting AI Agents

Scenario

We consider a system composed of:

- N AI agents
- Each agent runs a Small Language Model (SLM)
- Agents interact over an overlay graph $G = (V, E)$
  - Nodes V: AI agents
  - Edges E: communication channels

The system is **fully decentralized**

- System behavior is no longer local, but **emergent and topology-dependent**

Why this is relevant

Future AI infrastructures will be:

- Distributed
- Autonomous
- Cooperative
- Heterogeneous

Examples:

- Personal agents interaction
- Distributed fraud detection
- Autonomous vehicle swarms
- Cyber threat intelligence sharing
- Edge IoT infrastructures

Why this is relevant

Agent Interaction Model

Each agent maintains a state s_i^t

Update rule:

$$s_i^{t+1} = F_i\left(s_i^t, \{s_j^t : j \in \mathcal{N}(i)\}\right)$$

where:

- $\mathcal{N}(i)$ = neighbors of agent i
- F_i implemented via SLM reasoning

Security implication

Local compromise → Global instability

Threat 1: Compromised High-Centrality Agent

Assume node k has:

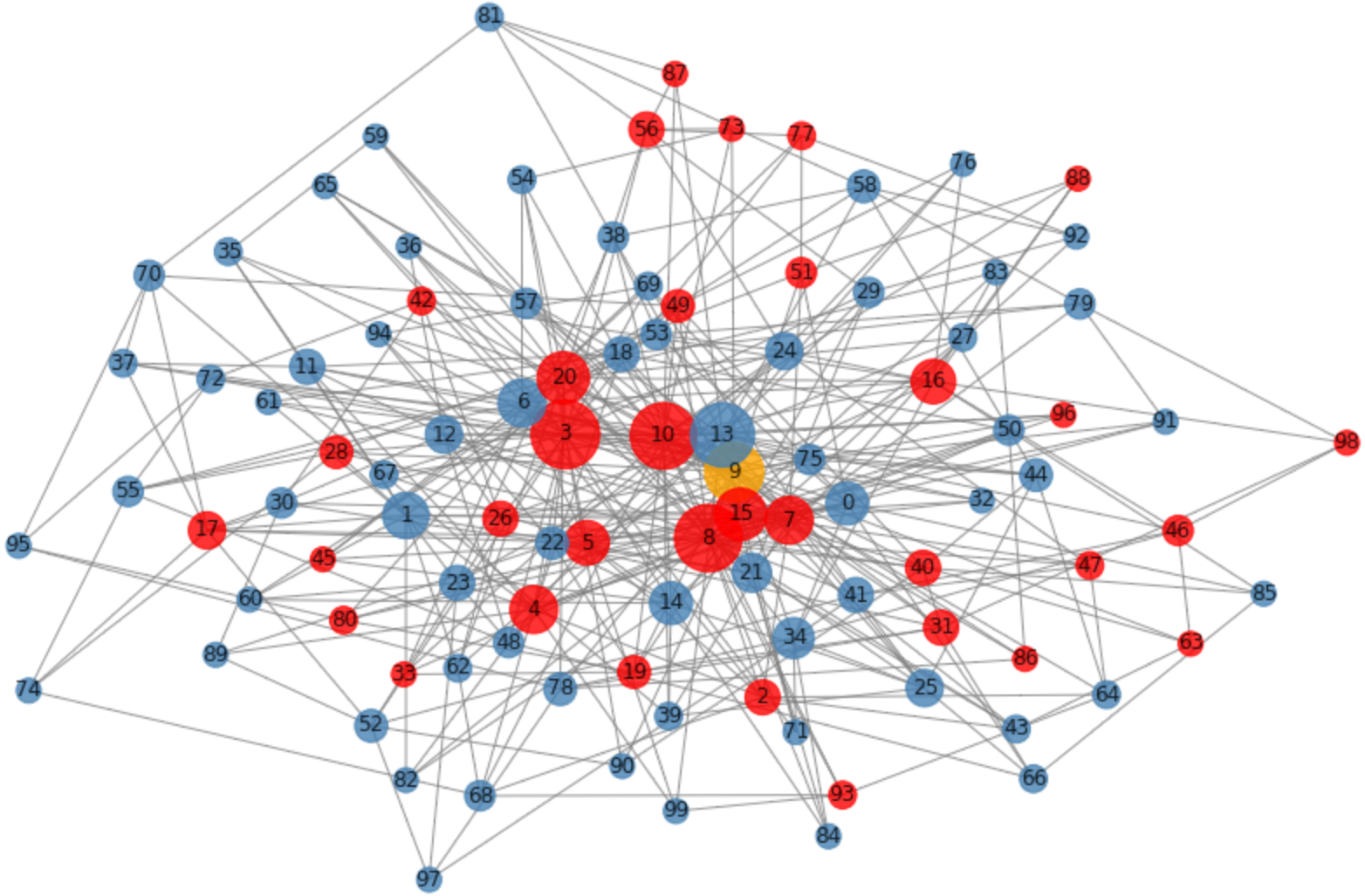
- High degree centrality
- High betweenness centrality
- High eigenvector centrality

If compromised: $s_k^t \rightarrow s_k^t + \delta$

Impact

- Rapid misinformation propagation
- Cascading behavioral drift
- System-wide instability

Threat 1: Compromised High-Centrality Agent



Threat 2: Coordinated Minority Attack

Subset $S \subset V$ behaves adversarially

Effects:

- Biased consensus
- Polarization
- Decision boundary shift

Critical question:

Is there a **phase transition threshold** for successful attack?

Threat 3: Topology-Driven Amplification

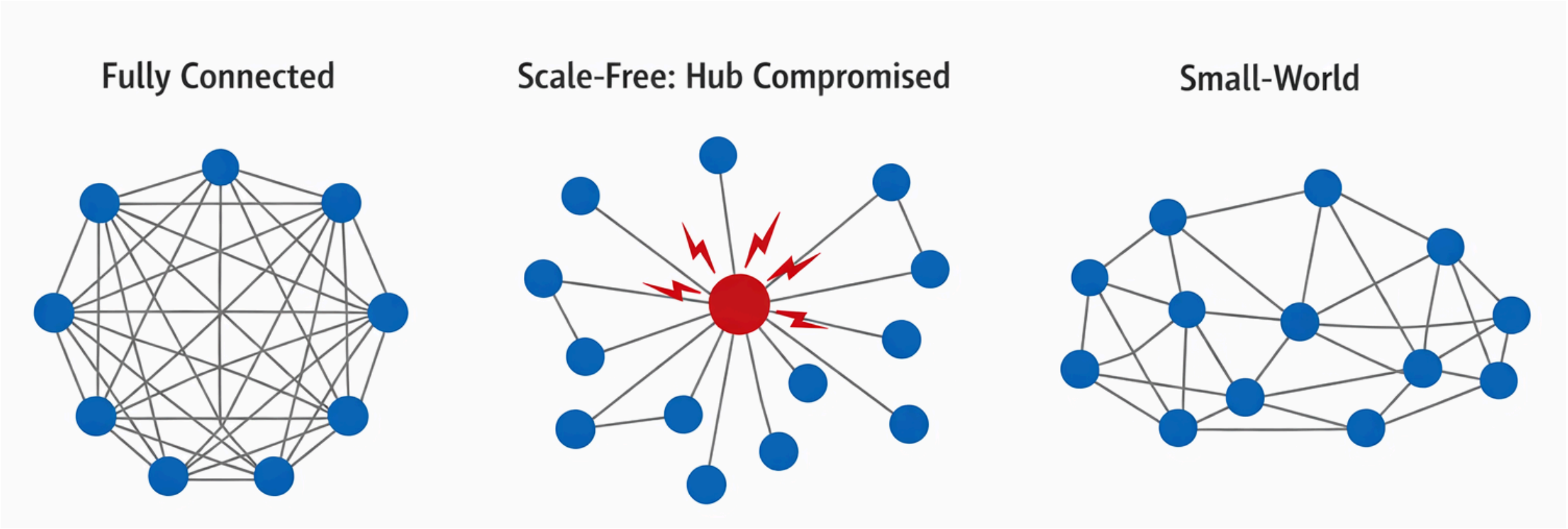
Certain graph structures amplify perturbations:

- Scale-free networks
- Highly clustered communities

Small perturbations may propagate exponentially

Security is topology-sensitive

Threat 3: Topology-Driven Amplification



Why Network Theory is Necessary

This is not a classical ML security problem, we are deadling with a **dynamical system over a graph**

We must analyze:

- Degree distribution
- Network connectivity
- Clustering coefficient
- Percolation thresholds

Mitigation 1: Trust-Weighted Aggregation

Modify update rule:

$$s_i^{t+1} = F_i\left(s_i^t, \sum_{j \in \mathcal{N}(i)} w_{ij} s_j^t\right)$$

Where:

- w_{ij} = trust weights
- dynamically updated

Goal: reduce adversarial influence

Mitigation 2: Topology Engineering

Design graph to improve robustness:

- Increase net connectivity
- Limit hub dominance
- Avoid extreme scale-free structures
- Introduce redundancy

In this scenario, security is partially a graph design problem

Mitigation 3: Byzantine-Resilient Consensus

Use robust aggregation:

- Median-based update
- Trimmed mean
- Outlier rejection

Goal: Bound adversarial impact even if fraction f of nodes is malicious

Mitigation 4: Influence Monitoring

Compute periodically:

- Node centrality metrics
- Influence functions
- Local divergence indicators

Early detection of:

- Behavioral drift
- Coordinated anomalies

Lessons Learned (AI Agent Networks)

What really matters

- Security is **emergent and topology-dependent**
- Local compromises can cause **global failures**
- Trust must be **distributed and dynamic**

What we need to study

- Network theory and graph robustness
- Distributed consensus under adversaries
- Trust and reputation systems
- Dynamical systems and stability analysis

Part III: Some problems and techniques we have identified so far

Foundational Terms - CIA

Foundational Terms - CIA + ML Extension

The classic **CIA triad** remains the backbone:

Property	Classic Definition	ML Extension
Confidentiality	Data visible only to authorized parties	Training data, model weights, inference outputs
Integrity	Data/system not modified without authorization	Model predictions not manipulated by adversary
Availability	System accessible when needed	Model not degraded by denial-of-service or poisoning

Threat Actors & Their Capabilities

We classify adversaries along two axes:

Knowledge:

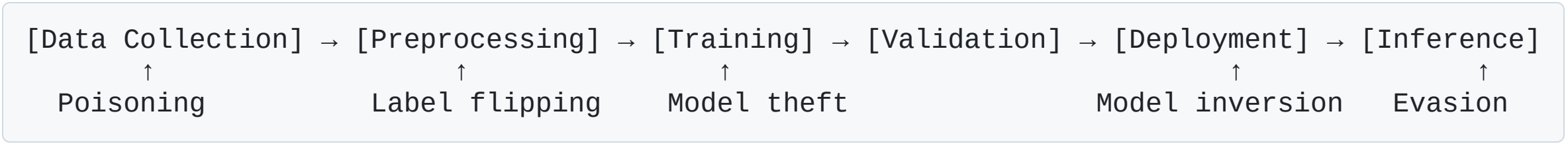
- **White-box**: full access to model architecture, weights, training data
- **Gray-box**: partial knowledge (architecture known, weights unknown)
- **Black-box**: query-only access (most realistic in production)

Goals:

- **Integrity attacks**: cause misclassification
- **Privacy attacks**: extract sensitive information from model
- **Availability attacks**: degrade model performance
- **Backdoor attacks**: embed hidden behavior triggered by a specific input pattern

Threat Modeling for ML Systems

The ML pipeline as an attack surface:



AI for Cybersecurity

to conclude ... since we gave it for granted

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AI as a Tool for Defense Cybersecurity

Application	ML Approach	Key Challenge
Intrusion Detection	Anomaly detection, sequence models (e.g., LSTM)	Maintaining low false positives under high-volume traffic
Malware Classification	Static/dynamic analysis, GNNs	Robustness to adversarial evasion techniques
Fraud Detection	Supervised learning (e.g., gradient boosting), online learning	Drift due to evolving attacker adaptation
Behavioral Analytics	Sequence modeling, user embeddings	Preserving user privacy while modeling behavior
Threat Intelligence	NLP on CTI reports, knowledge graphs	Data quality and noise in OSINT

Some Recommended Resources to Start

Primary references:

- Goodfellow et al. — *Deep Learning* (MIT Press) — Ch. on Robustness
- Dwork & Roth — *The Algorithmic Foundations of Differential Privacy* (2014)
- Papernot et al. — *SoK: Security and Privacy in ML* (IEEE EuroS&P 2018)

Tools & frameworks:

- `CleverHans`, `ART (IBM)` — adversarial attack/defense libraries
- `Opacus` (PyTorch), `TF Privacy` — differential privacy
- `PySyft` — federated learning & SMPC
- `FATE` — industrial federated learning framework

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